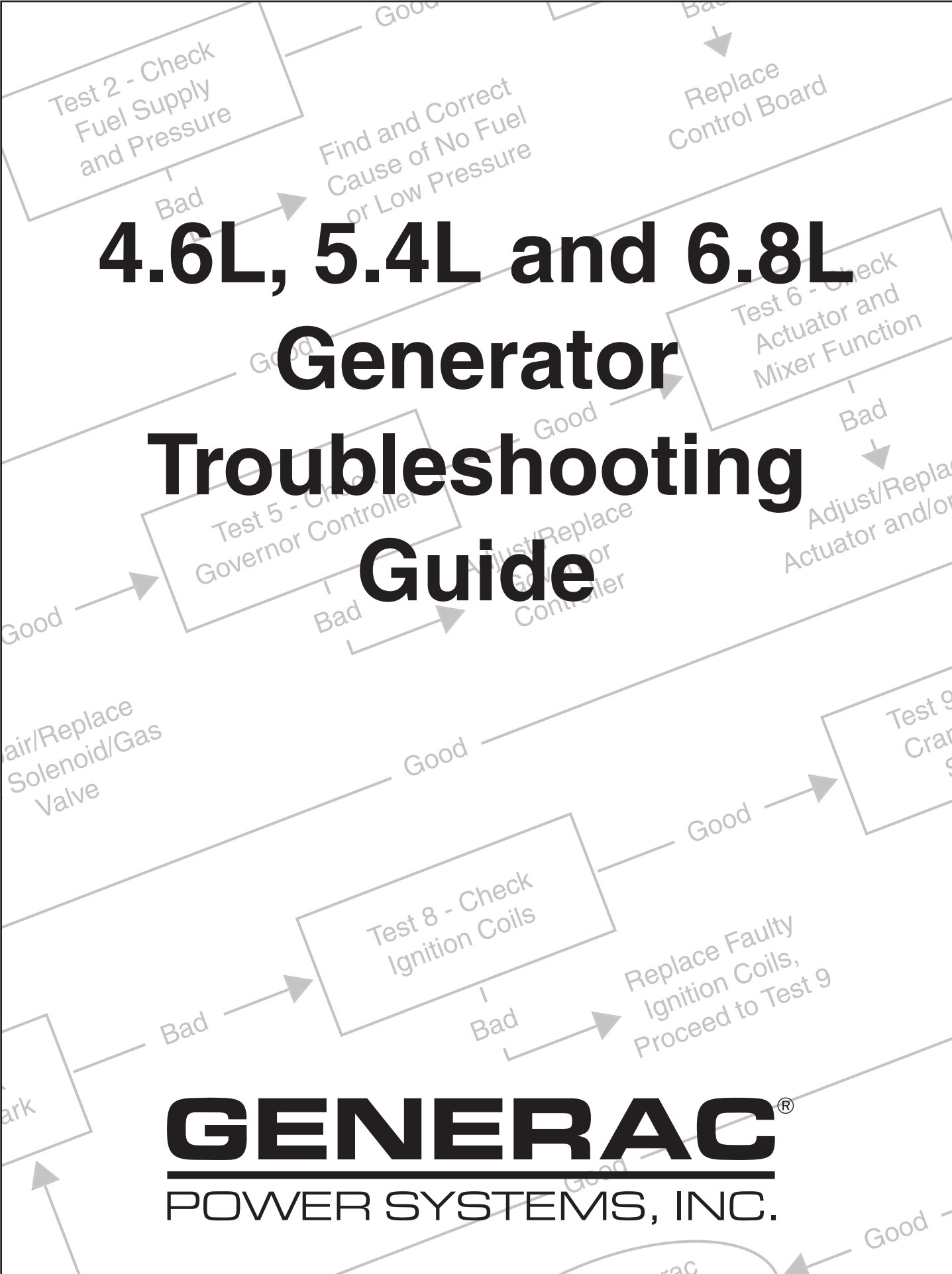




The background of the page features a faint, tilted flowchart for troubleshooting generator issues. The flowchart includes several test steps in rectangular boxes: 'Test 2 - Check Fuel Supply and Pressure', 'Test 5 - Check Governor Controller', 'Test 6 - Check Actuator and Mixer Function', 'Test 8 - Check Ignition Coils', and 'Test 9 - Check Crankshaft Position'. Arrows indicate the flow between these tests, with labels 'Good' and 'Bad' indicating the outcome of each test. For example, a 'Bad' result from Test 2 leads to 'Find and Correct Cause of No Fuel or Low Pressure', while a 'Good' result leads to 'Test 5'. A 'Bad' result from Test 5 leads to 'Adjust/Replace Governor Controller', and a 'Good' result leads to 'Test 8'. A 'Bad' result from Test 8 leads to 'Replace Faulty Ignition Coils, Proceed to Test 9', while a 'Good' result leads to 'Test 9'. A 'Bad' result from Test 6 leads to 'Adjust/Replace Actuator and/or Mixer', and a 'Good' result leads to 'Test 9'. A 'Bad' result from Test 9 leads to 'Replace Control Board'. The flowchart is partially obscured by the main title text.

4.6L, 5.4L and 6.8L Generator Troubleshooting Guide

GENERAC[®]
POWER SYSTEMS, INC.

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INTRODUCTION

Perform the “Diagnostic Tests” in conjunction with the “Troubleshooting Flow Charts”.

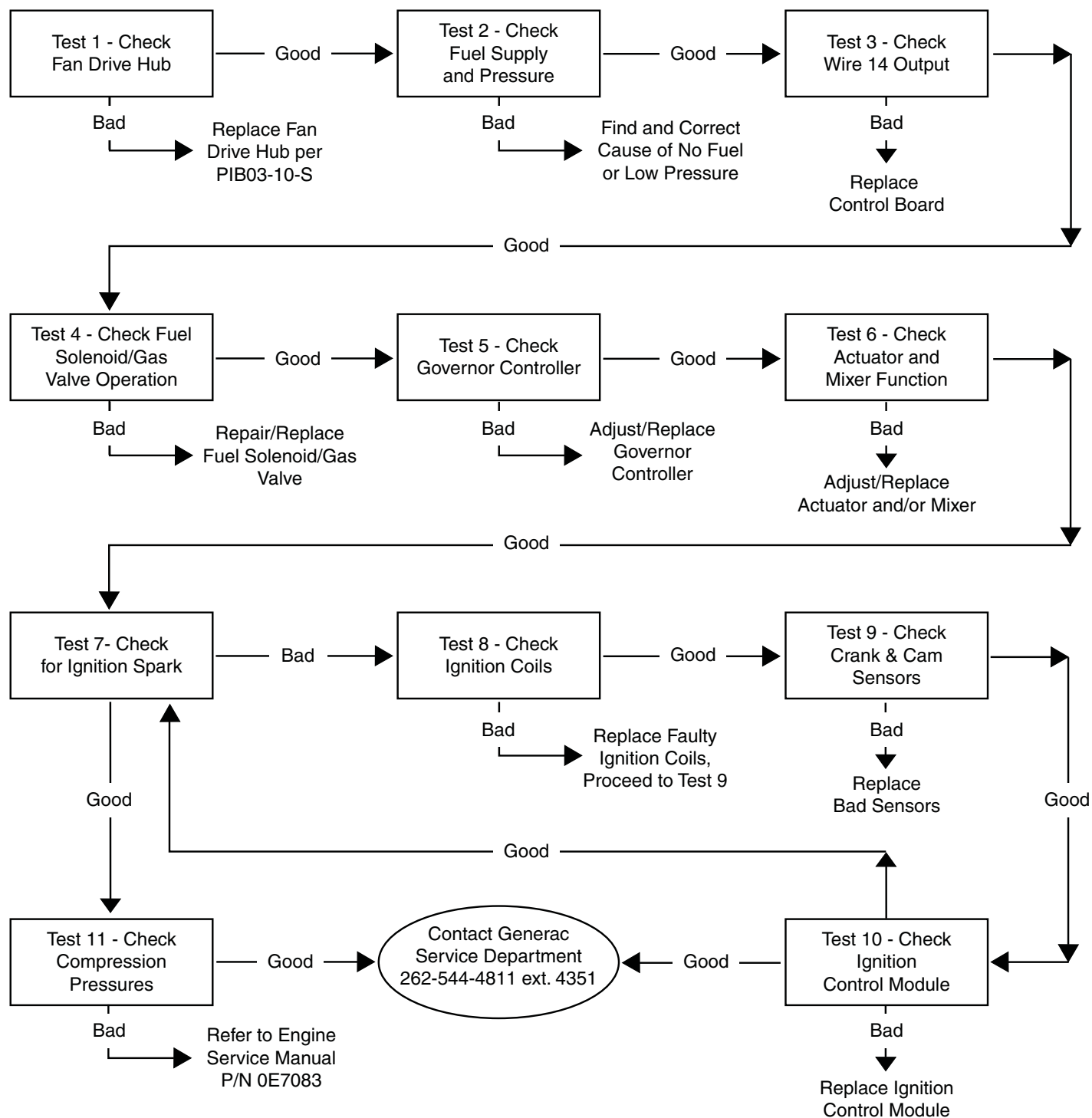
The test procedures and methods presented in this guide are not exhaustive. We could not possibly know of, evaluate and advise the service trade of all conceivable ways in which testing and trouble diagnosis might be performed. We have not undertaken any such broad evaluation.

NOTE:

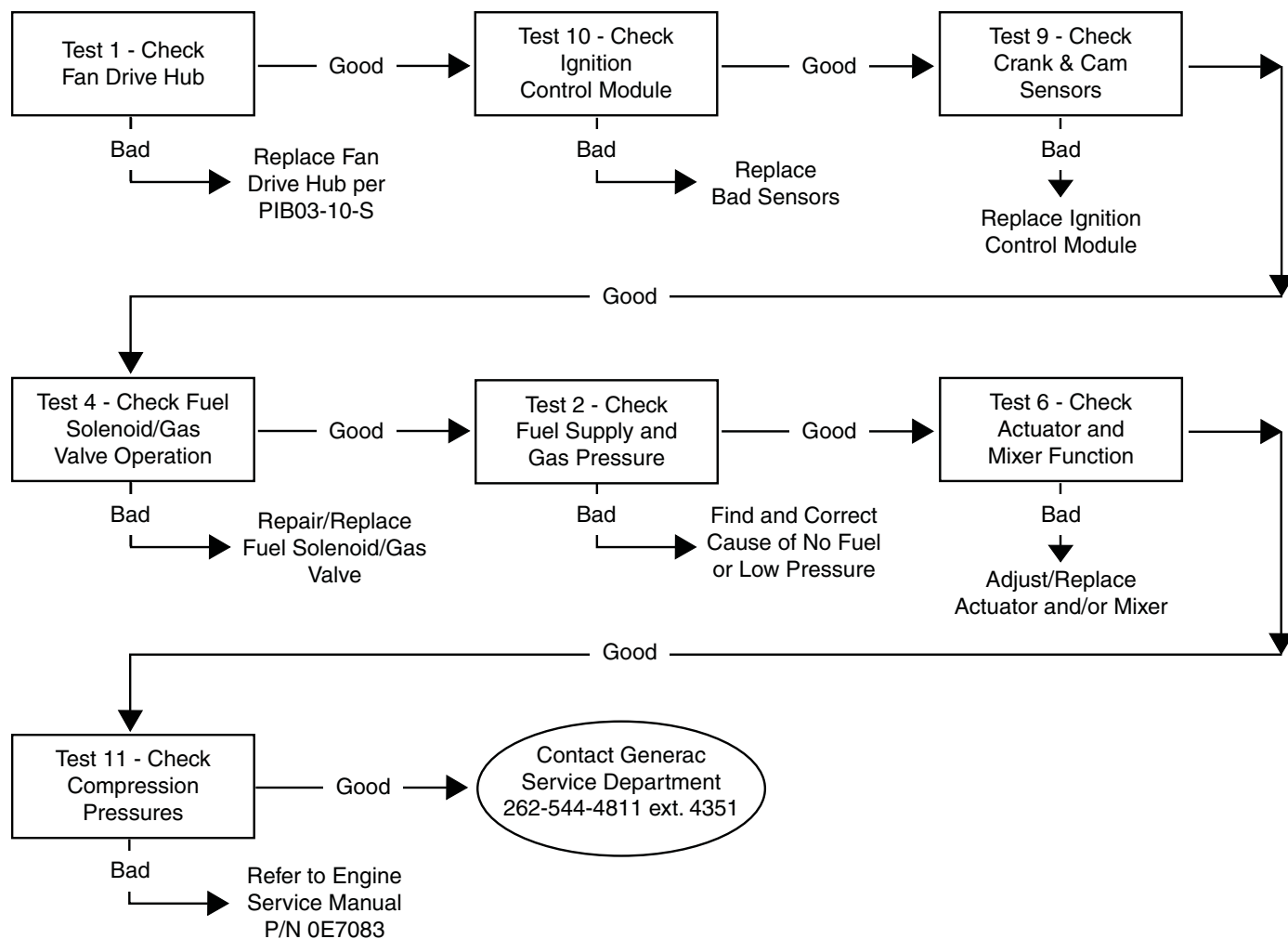
Do not force meter leads into any housing or electrical connection, to do so may damage the housing or the electrical connection. Refer to PIB05-14-S for recommended wire removal, replacement and testing options.

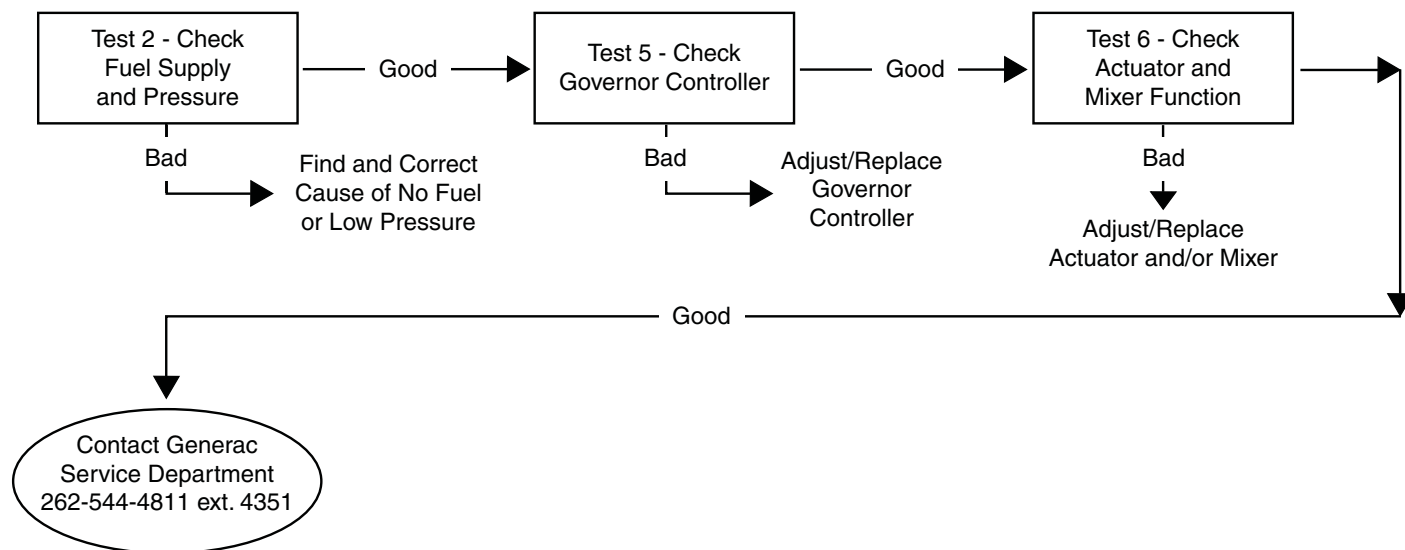
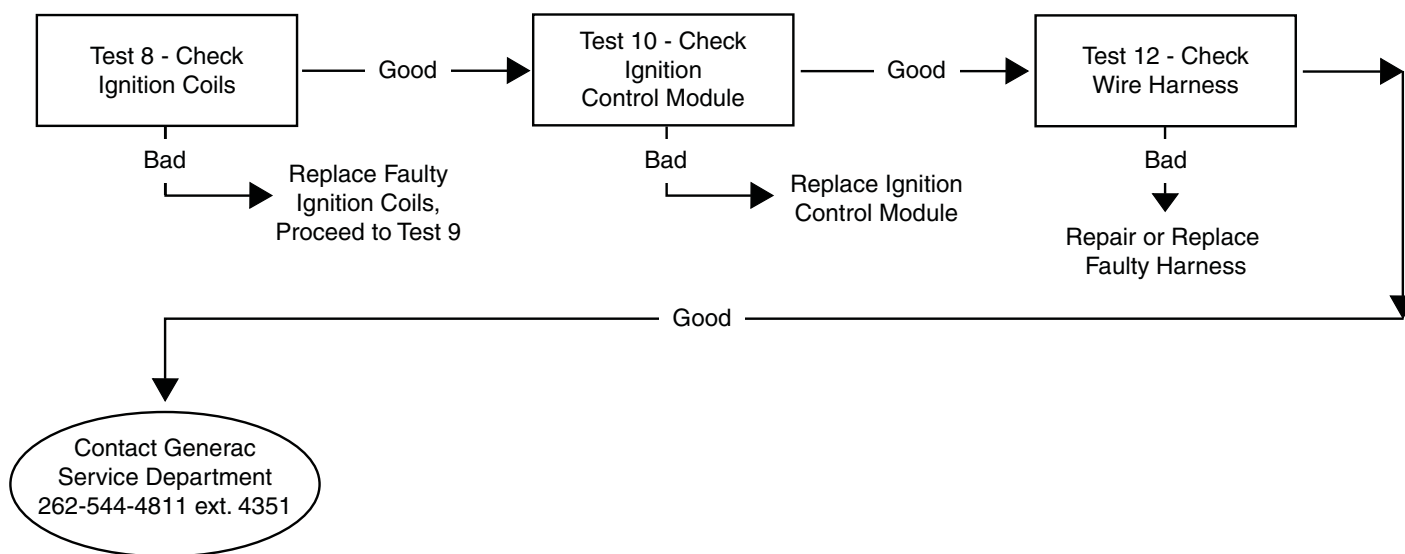
For additional information, reference the following manuals:

- Part no. 0E7083; 6.8L Engine Service Manual
- Part no. 0E6478; 6.8L Engine Parts Manual

PROBLEM 1 - UNIT CRANKS BUT WILL NOT START

PROBLEM 2 - UNIT CRANKS AND BACKFIRES



PROBLEM 3 - UNIT STARTS HARD AND RUNS ROUGH/LACKS POWER OR HUNTS**PROBLEM 4 - 10 AMP IN-LINE FUSE (F1) BLOWN**

TEST 1 – CHECK FAN DRIVE HUB

◆ DISCUSSION

Gear driven units manufactured prior to 10/16/03 had a fan drive hub installed that could possibly fail and should have been replaced according to PIB03-10-S. To verify the correct fan drive hub, go onto service OIS and enter the unit S/N, look under “Engine and Accessories” for P/N OE8909, if it is not there, contact the Service Department and we can verify if PIB03-10-S was performed.

⚠ CAUTION ⚠

⚠ To prevent injury, perform the following steps in the order indicated before proceeding with the following procedure:

- Disable the generator set from starting by setting the Auto/Off/Manual switch to the “OFF” position.
- Remove the control panel fuse.
- Turn off the battery charger.
- Remove the negative battery cable.

◆ PROCEDURE

Follow the procedure given in PIB03-10-S, which is attached to the back of this bulletin.

◆ RESULTS

If, after replacing the fan drive hub and the unit still will not start, proceed to “TEST 11 – Check Compression Pressures” before continuing through the “Troubleshooting Flow Charts”.

TEST 2 – CHECK FUEL SUPPLY AND PRESSURE

◆ DISCUSSION

The engine-generator set was factory tested and adjusted using the primary fuel source (natural gas, LP vapor or LP liquid) as the fuel supply.

- An adequate gas supply and sufficient fuel pressure must be available or the engine will not start.
- Minimum recommended gaseous fuel pressure at the generator fuel inlet connection is 7 inches water column (4.05 ounces per square inch).
- Maximum gaseous fuel pressure at the generator fuel inlet connection is 14 inches water column (8.09 ounces per square inch).
- The gaseous fuel system must be properly tested for leaks following installation and periodically thereafter. No leakage is permitted. Leak test methods must comply strictly with gas codes.

⚠ DANGER ⚠

⚠ GASEOUS FUELS ARE HIGHLY EXPLOSIVE. DO NOT USE FLAME OR HEAT TO TEST THE FUEL SYSTEM FOR LEAKS. NATURAL GAS IS LIGHTER THAN AIR AND TENDS TO SETTLE IN HIGH PLACES. LP GAS IS HEAVIER THAN AIR AND TENDS TO SETTLE IN LOW AREAS. EVEN THE SLIGHTEST SPARK CAN IGNITE THESE GASES AND CAUSE AN EXPLOSION.

◆ PROCEDURE

A water manometer or a gauge that is calibrated in “ounces per square inch” may be used to measure the fuel pressure. Fuel pressure at the inlet side of the fuel solenoid valve should be between 7 –14 inches water column as measured with a manometer, or 4.058 - 8.09 ounces per square inch as measured with a pressure gauge.

The fuel pressure can be checked using a pressure tester kit (P/N OC7977).

NOTE:

Where a primary regulator is used to establish fuel inlet pressure, adjustment of that regulator is usually the responsibility of the fuel supplier or the fuel supply system installer.

◆ RESULTS

1. If fuel supply and pressure are adequate, but engine will not start, continue following the “Troubleshooting Flow Charts”.
2. If generator starts but runs rough or lacks power, repeat the above procedure with the generator running and under load. The fuel system must be able to maintain 7” – 14” water column at all load requirements. If proper fuel supply and pressure is maintained, go to Test 5.

TEST 3 – CHECK WIRE 14 OUTPUT

◆ DISCUSSION

C Option Control Board and E Option Control Module

During any cranking action, the controller’s crank output (Wire 56 or 56A) and run output (Wire 14 or 14A) both energize simultaneously. When the run circuit energizes, 12 volts DC is delivered to Wire 14 (or Wire 14A) and to the fuel solenoid. The solenoid energizes open to allow fuel flow to the engine. This test will determine if the circuit is working properly.

H-100 Controller and G-100 Controller

During any cranking action, the controller's crank output (Wire 56A) and run output (Wire 256) both close to provide a ground path. When the run circuit energizes, 12 volts DC is delivered to Wire 14 through a relay contact when the coil is energized by the grounding of wire 256 on RB1, which delivers DC voltage to the fuel solenoid. The solenoid energizes open to allow fuel flow to the engine. This test will determine if the circuit is working properly.

NOTE:

Do not force meter leads into any housing or electrical connection, to do so may damage the housing or the electrical connection. Refer to PIB05-14-S for recommended wire removal, replacement and testing options.

◆ PROCEDURE

1. Set the AUTO/OFF/MANUAL switch to OFF.
2. C Option Control Board and E Option Control Module: Connect the positive (+) test lead of a DC voltmeter (or VOM) onto Wire 14 (or Wire 14A) of the control circuit board connector. Connect the common (-) test lead to frame ground. The meter should be set to DC Volts. H-100 Controller and G-100 Controller: Connect the common (-) test lead of a DC voltmeter (or VOM) onto Wire 256 of the controller or the relay board connector. Connect the positive (+) test lead to Wire 15 of the relay board connector. The meter should be set to DC Volts.
3. While observing the meter, set the AUTO/OFF/MANUAL switch to MANUAL.
 - a. The control circuit's crank and run outputs should energize and the engine should crank and start.
 - b. The meter should indicate battery voltage.
 - c. If battery voltage is indicated, proceed to Step 4. If battery voltage is NOT indicated, proceed to Test 3 results.
4. Disconnect Wire 14 (or Wire 14A) at the fuel solenoid (FS).
 - a. Connect the positive (+) test lead to Wire 14 (or Wire 14A). Connect the negative (-) test lead to a clean frame ground. Set the AUTO/OFF/MANUAL switch to MANUAL. Battery voltage should be measured. If battery voltage is indicated, proceed to Step 5.
 - b. Connect the positive (+) test lead to Wire 14 (or Wire 14A) at the TB1 terminal strip (C Option Control Panel) or at the BC3 connector pin 9 (E Option Control Panel) or at the relay board (H-100 and G-100 Controllers). Connect the negative (-) test lead to frame ground. While observing the meter, set the AUTO/OFF/MANUAL switch to MANUAL. Battery voltage should be measured.

5. Set the VOM to it's "R x 1" scale.
6. Disconnect Wire 0 from the Fuel Solenoid (FS). Connect one test lead to Wire 0 and the other test lead to a clean frame ground. CONTINUITY should be measured.

◆ RESULTS

1. If the engine cranks but does not start and battery voltage was NOT measured in Step 3, replace either the C Option Control Board or the E Option Control Module, whichever type is installed on the unit.
2. If the engine cranks and battery voltage was measured in Step 3, but there was no battery voltage in Step 4 (a), repair or replace Wire 14 or Wire 14A between the Fuel Solenoid (FS) and either the TB1 terminal strip or the BC3 connector.
3. If the engine cranks and battery voltage was measured in Step 3 and no battery voltage is measured in Step 4(b), repair or replace Wire 14 or Wire 14A between the connector on the control board and the either the TB1 terminal strip or the BC3 connector.
4. If the engine cranks but does not start and battery voltage was measured in Steps 2, 3, 4, and CONTINUITY was NOT measured in Step 6, proceed to Test 4.

TEST 4 – CHECK FUEL SOLENOID

◆ DISCUSSION

In Test 3, if battery voltage was delivered to Wire 14 or Wire 14A, the fuel solenoid should have energized open. This test will verify whether the fuel solenoid is operating.

◆ PROCEDURE

1. Disconnect Wire 56 or Wire 56A from the starter contactor relay. This will disable the unit from cranking.
2. Place one hand on the Fuel Solenoid (FS). Cycle the AUTO/OFF/MANUAL switch from OFF to MANUAL. You should be able to feel the solenoid actuate as well as hear it actuate.
3. Set the AUTO/OFF/MANUAL switch to OFF. Reconnect Wire 56 or Wire 56A to the starter contactor relay.

◆ RESULTS

1. If the solenoid actuates, go to Test 5.
2. Replace the solenoid if it does not actuate.

TEST 5 – CHECK GOVERNOR CONTROLLER

◆ DISCUSSION

In the previous tests, we verified that fuel is available to the engine. If the governor controller is not functioning properly, the engine will not get the right amount of air and fuel for proper combustion.

The two most common governor controllers that have been used on the 6.8 liter engine are the Barber Coleman Dyn1 and the Woodward DPG-2201-001.

The first two steps of the following procedure will verify that the governor controller inputs are within acceptable ranges. Step three of the procedure is for the DPG-2201 controller only, and steps four, five and six in the procedure are for the Dyn1 controller only.

◆ PROCEDURE

1. Check battery voltage at terminals 1 and 2 on the governor controller. Voltage reading should be within +/- 20% of nominal supply voltage.
2. Check the magnetic pickup signal to the controller. Measure the AC voltage across terminals 10 and 11 (Dyn1 controller) or across terminals 5 and 6 (DPG-2201 controller) while cranking the engine. Voltage should be a minimum of 2.5VAC for the Dyn1 controller, and 2.0VAC for the DPG-2201 controller.
3. (DPG-2201 only) Check the actuator output duty cycle. There are two methods to determine a good value to use for this parameter.

Method 1: Use a meter to measure the duty cycle across the ACT terminals of the governor while the engine is running. **Note:** To determine if your meter is reading the correct value:

- a) First, apply power to the governor but do NOT start the engine.
- b) Second, measure the duty cycle across the ACT terminals (#3 and #4). The reading should be 5 for 5%. If the reading is 95, then reverse the leads.

Method 2: Use a PC running the Universal PST application (which can be downloaded at the following web address <http://www.woodward.com/software/software.cfm>, and the manual at <http://www.woodward.com/pdf/ic/36523.pdf>) and read the value of the PWM command in the View Status panel when the engine is cranking. Note: from the Universal PST's startup screen press the View Status button, then press the Start Monitoring button to begin reading values.

4. With DC to the governor ON and the engine OFF, measure the DC voltage from terminal 6 (+) to terminal 2 (-).

5. Measure the DC voltage between terminal 7 (+) to terminal 2 (-).
6. With DC power OFF remove the leads at the actuator. Check the continuity of each terminal to the case.

◆ RESULTS

1. If the voltage measured at terminals 1 and 2 on the controller was not within the +/- 20% of nominal supply voltage, check battery and connections, repair or replace as required.
2. If the AC voltage measured in the test was below the minimum in step 2, adjust or replace the magnetic pickup as required.
3. If the duty cycle measured in step 3 was not at least 5%, then restore the factory default setting and repeat the test.
4. The voltage reading should be approx. 8 VDC, if it is not, replace the controller.
5. The voltage reading should be approx. 4 VDC, if it is not, replace the controller.
6. If continuity is detected, replace the controller.

TEST 5A – CHECK GOVERNOR DRIVER (H-100/G-100 PRODUCT LINE)

◆ DISCUSSION

The governor on the H-100/G-100 product line is built into the controller and uses PID settings to control the engine much in the same manner as an external governor controller. There is a governor driver board that takes the PWM signal from the H Panel output and converts that signal to a variable DC signal to drive the actuator.

NOTE:

Do not force meter leads into any housing or electrical connection, to do so may damage the housing or the electrical connection. Refer to PIB05-14-S for recommended wire removal, replacement and testing options.

◆ PROCEDURE

1. Check battery voltage at terminals RB1-7 (Wire 14) and RB1-3 (Wire 0) while the engine is cranking or running. Voltage reading should be within + / - 20% of nominal supply voltage.
2. Check the magnetic pickup signal to the H Panel controller. Measure the AC voltage across terminals J1-24 (Wire 0) and J1-25 (Wire 79) while cranking the engine. Voltage should be a minimum of 0.5VAC. Additionally, using Genlink you can verify that the H-Panel is seeing the MPU input by checking the RPM reading on the main display.

3. Check the controller output duty cycle to the governor driver module. Use a meter to measure the duty cycle across the governor driver connector pin 12 (Wire 769) and pin 10 (Wire 0) while cranking the engine. The reading should be 80 for 80%. If the reading is 20, then reverse the leads.

◆ RESULTS

1. If the voltage measured at terminals 1 and 2 on the controller was not within the +/- 20% of nominal supply voltage, check battery and connections, repair or replace as required.
2. If the AC voltage measured in the test was below the minimum in step 2, adjust or replace the magnetic pickup as required.
3. If the duty cycle measured in Step 3 was not at least 80%, contact the Service Department at 1-262-544-4811 ext.4351.

TEST 6 – CHECK ACTUATOR AND MIXER FUNCTION

◆ DISCUSSION

The actuator is controlled by the governor controller or the governor driver module. The opening of the mixer (configured units) is controlled by the amount of engine vacuum produced during cranking.

◆ PROCEDURE

1. Verify that there is no binding throughout the full range of motion of the actuator shaft and linkage on units with a PowerFlow actuator.
2. Disconnect the wire harness from the Bosch Actuator.
 - Measure resistance across Pin 6 & Pin 3, and measure resistance across Pin 6 & Pin 2 while the actuator is in the “at rest” position.
 - Measure resistance across Pin 6 & Pin 3, and measure resistance across Pin 6 & Pin 2 while the actuator is in the “full open” position.
3. Verify the mixer is opening on configured units. While cranking the engine, lift the diaphragm by hand and see if the engine starts.

◆ RESULTS

1. If there is any binding in the linkage, repair or replace as required.
2. At Rest: Pin 6 & Pin 3 – 1.4K Ohms
Pin 6 & Pin 2 – 700 Ohms
Full Open: Pin 6 & Pin 3 – 600 Ohms
Pin 6 & Pin 2 – 1.4K Ohms
3. If the engine starts when the diaphragm is lifted, proceed to TEST 11 – CHECK COMPRESSION PRESSURES. If compression pressures are fine, replace the mixer assembly.

TEST 7 – CHECK FOR IGNITION SPARK

◆ DISCUSSION

Previously, we verified that the engine should be getting the correct mixture of air and fuel for the combustion process, next we will check to see if the engine is also getting spark in order to ignite the air and fuel mixture.

◆ PROCEDURE

The 4.6L, 5.4L and 6.8L engines uses a Coil On Plug (COP) design, which is a separate coil per spark plug and each coil is mounted directly onto the plug.

- Shut off the gas supply to the unit.
- Remove the ignition coil and attach a spark tester (P/N 0C5969) to the end of the ignition coil boot and to a good ground point.
- Crank the engine and observe the spark tester, make note of the spark condition for the cylinder.
- Remove and inspect the spark plug, verify that the gap is set to 0.054 inches. Reinstall the spark plug.
- Repeat the previous steps for all cylinders.

◆ RESULTS

- If weak spark or no spark was noted, proceed with TEST 8 – CHECK IGNITION COILS.
- If a good spark was noted, proceed with TEST 11 – CHECK COMPRESSION PRESSURES.

TEST 8 – CHECK IGNITION COILS

◆ DISCUSSION

A weak spark can be caused by ignition coils insulation breaking down and providing an alternate path to ground. A no spark condition can be caused by an ignition coil being in an open circuit condition or a direct short condition, internal winding to winding or to ground.

The ICM acts as an electronic switch to ground in the coil primary circuit. When the switch is closed battery positive voltage (Wire 15A) or (Wire 15D) applied to the coil primary circuit builds a magnetic field around the primary coil. When the switch opens, the power is interrupted and the primary field collapses inducing the high voltage in the secondary coil windings and the spark plug is fired.

NOTE:

Do not force meter leads into any housing or electrical connection, to do so may damage the housing or the electrical connection. Refer to PIB05-14-S for recommended wire removal, replacement and testing options.

◆ PROCEDURE

1. Check the ignition coil resistances across the terminals and to ground. Resistance readings across the terminals should be between 0.3 ohms and 0.8 ohms. Resistance readings between each terminal and ground should be infinite.
2. Check for battery voltage at terminal 1 or B+ (Wire 15A or Wire 15D) and ground. Measured voltage should be at least 10 volts DC.
3. Connect a meter set to read frequency across the primary terminals of the coil and crank the engine.

◆ RESULTS

1. If any of the coil resistances measured were not within the range specified, or there was any reading to ground, replace the suspect ignition coil(s).
2. If the voltage measured in Step 2 is not within the range specified, recharge or replace the battery. If no voltage was measured in Step 2, proceed to Test 10 – Check Ignition Control Module.
3. The frequency reading should be one-half of the engine rotation speed during cranking. If no frequency is indicated, proceed to TEST 10 – CHECK IGNITION CONTROL MODULE.

TEST 9 – CHECK CRANK AND CAM SENSORS

◆ DISCUSSION

The camshaft position sensor (CMPS) identifies when piston #1 is on the compression stroke. The ICM uses this information to synchronize the firing of the individual coils. The crankshaft position sensor (CKPS) is the primary sensor for ignition information to the ICM. The CKPS input is used by the ICM to determine engine speed and angular position of the crankshaft. The location of the crankshaft position sensor is shown in figure 9.1. The position of the camshaft position sensor is shown in figure 9.2.

Figure 9.1

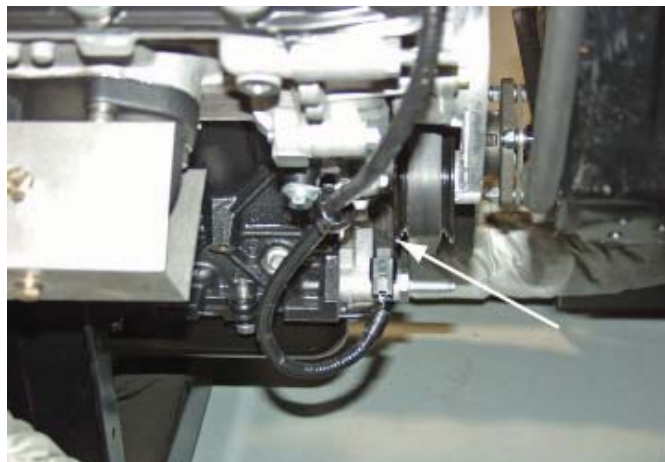


Figure 9.2



NOTE:

Do not force meter leads into any housing or electrical connection, to do so may damage the housing or the electrical connection. Refer to PIB05-14-S for recommended wire removal, replacement and testing options.

◆ PROCEDURE

1. Measure the resistance of the CKPS. Disconnect the 2-pin connector from the CKPS and measure the resistance across pins 1 & 2 of the CKPS. The resistance measured should be between 290 – 390 ohms.
2. Measure the resistance of the CMPS. Disconnect the 2-pin connector from the CMPS and measure the resistance across pins 1 & 2 of the CMPS. The resistance measured should be between 290 – 390 ohms.

◆ RESULTS

1. If the resistance measured in Step 1 does not fall within the tolerances stated, replace the CKPS (P/N OD34540231).
2. If the resistance measured in Step 2 does not fall within the tolerances stated, replace the CMPS (P/N OD34540230).

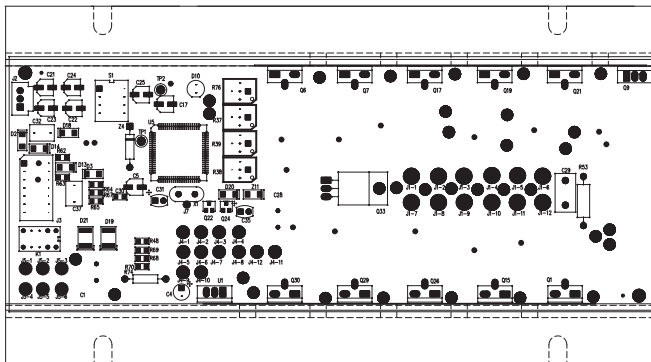
TEST 10 – CHECK IGNITION CONTROL MODULE

◆ DISCUSSION

The ICM controls all aspects of the engine ignition based on the inputs it receives from the CMPS and the CKPS. In the previous tests we checked the inputs to the ICM and verified the integrity of the ignition coils.

The default settings for the ignition modules on configured units without the H-100/G-100 controllers are as shown in Figure 10.1 (5.4L and 6.8L) and Figure 10.2 (4.6L). Please note that the position of the DIP switch "14 LINE" will be determined by whether or not wire 14 is connected to terminal 12 of the power connector on the ignition control module. **Dip switches are factory set in the "ON" (up) position (Figure 10.3).**

Figure 10.1



R76(AIR FUEL)
AIR/FUEL = 3: 0' CLOCK

R37(D)
GOV DIFF = 1: 0' CLOCK

R39(P)
GOV GAIN = 1: 0' CLOCK

R38(I)
GOV STAB = 12: 0' CLOCK

S1 (DIP SWITCH)
18 • ON
23 • OFF
30 • OFF
14 LINE • OFF

Figure 10.2

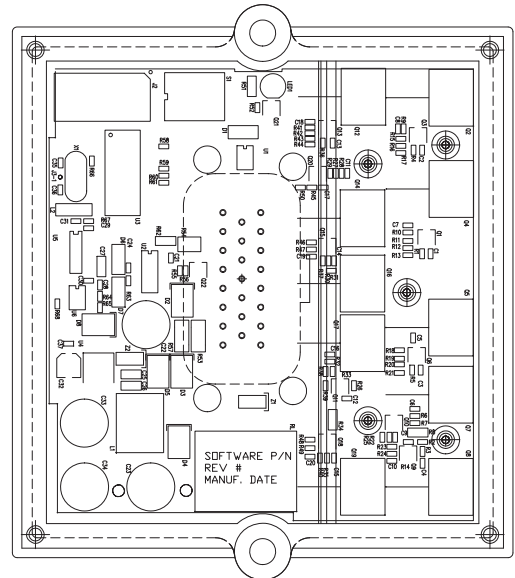
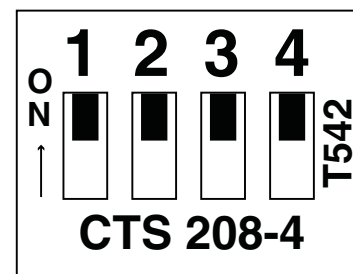


Figure 10.3



NOTE:

Electronic Ignition engine timing is entirely controlled by the ICM. Electronic Ignition engine timing is NOT adjustable. Do not attempt to check base timing, you will receive false readings.

Ignition Control Modules (P/N: 0F8656 for the 4.6L, and P/N: 0F3113 for the 5.4L and 6.8L) used with the H-100/G-100 product line have a red diagnostic LED inside the ignition module (that is visible without removing the ignition cover, but you will have to remove the ignition module so you can see the back of the module) that will indicate if the ignition has been energized. This red LED will flash at a 0.5 second ON and a 0.5 second OFF rate when the ignition is energized. This is considered (1) blink.

NOTE:

Do not force meter leads into any housing or electrical connection, to do so may damage the housing or the electrical connection. Refer to PIB05-14-S for recommended wire removal, replacement and testing options.

◆ PROCEDURE

1. Observe the red diagnostic LED on the ignition control module used on the H-100/G-100 product line. In the event that an ignition fault has occurred, the ignition will wait 60 seconds before powering down.
2. Measure voltage between Wire 56 (Pin 11 on the H-100/G-100 product line and Pin 3 of the 12-Pin power connector on other configured units) and Wire 0B (Pin 17 on the H-100/G-100 product line and Pin 2 of the 12-Pin power connector on other configured units). Battery voltage should be measured.
3. Measure voltage between Wire 15C (Pin 16 on the H-100/G-100 product line and Pin 1 of the 12-Pin power connector on other configured units) and Wire 0B (Pin 17 on the H-100/G-100 product line and Pin 2 of the 12-Pin power connector on other configured units). Battery voltage should be measured.
4. Measure voltage between Wire 14 (Pin 12 on the H-100/G-100 lines and Pin 12 of the 12-Pin power connector on other configured units) and Wire 0B (Pin 17 on the H-100/G-100 line and Pin 2 of the 12-Pin power connector on other configured units). Battery voltage should be measured.

◆ RESULTS

1. LED Fault Codes:
 - No Crank Signal: LED blinks 2 times, is OFF for 3.0 seconds and then repeats.
 - No Cam Signal: LED blinks 3 times, is OFF for 3.0 seconds and then repeats.

Only one fault is displayed at a time. If multiple faults exist then the highest priority fault must be resolved prior to a lower priority fault being displayed. If either fault exists, proceed to Test 12 – Check Wire Harness.

2. If battery voltage was not measured in Step 2, proceed to Test 12 – Check Wire Harness.
3. If battery voltage was not measured in Step 3, proceed to Test 12 – Check Wire Harness.
4. If battery voltage was not measured in Step 4, proceed to Test 12 – Check Wire Harness.

TEST 11 – CHECK COMPRESSION PRESSURES

◆ DISCUSSION

Low compression pressures inside the combustion chambers can make it very difficult to start an engine. Low compression pressures can be caused by faulty or worn piston rings, valves sticking or seating improperly, improper valve timing, or a leaking head gasket would be the primary reasons for low compression pressures.

◆ PROCEDURE

1. Make sure the oil in the crankcase is of the correct viscosity and at the proper level and that the battery is properly charged. Operate the unit until the engine is at normal operating temperature. Disable the unit from starting, shut off the gas supply to the unit and then remove all of the spark plugs.
2. Install a compression tester in the No. 1 cylinder.
3. Crank the engine a minimum of five compression strokes and record the highest reading. Note the approximate number of compression strokes required to obtain the highest reading.
4. Repeat the test on each cylinder, cranking the engine approximately the same number of compression strokes.

◆ RESULTS

The indicated compression pressures are considered within specification if the lowest reading cylinder is within 75 percent of the highest reading. Refer to the Compression Pressure Limit Chart.

If one or more cylinders read low, squirt approximately one tablespoon of clean engine oil on top of the pistons in the low-reading cylinders. Repeat the compression pressure check on these cylinders.

◆ **COMPRESSION PRESSURE LIMIT CHART**

MAX-MIN kPa (psi)	MAX-MIN kPa (psi)	MAX-MIN kPa (psi)	MAX-MIN kPa (psi)
924 - 696 (134 - 101)	1131 - 848 (164 - 123)	1338 - 1000 (194 - 146)	1544 - 1158 (224 - 168)
938 - 703 (136 - 102)	1145 - 855 (166 - 124)	1351 - 1014 (196 - 147)	1558 - 1165 (226 - 169)
952 - 717 (138 - 104)	1158 - 869 (168 - 126)	1365 - 1020 (198 - 148)	1572 - 1179 (228 - 171)
965 - 724 (140 - 106)	1172 - 876 (170 - 127)	1379 - 1034 (200 - 150)	1586 - 1186 (230 - 172)
979 - 738 (142 - 107)	1186 - 889 (172 - 129)	1393 - 1041 (202 - 151)	1600 - 1200 (232 - 174)
933 - 745 (144 - 109)	1200 - 903 (174 - 131)	1407 - 1055 (204 - 153)	1613 - 1207 (234 - 175)
1007 - 758 (146 - 110)	1214 - 910 (176 - 132)	1420 - 1062 (206 - 154)	1627 - 1220 (236 - 177)
1020 - 765 (148 - 111)	1227 - 917 (178 - 133)	1434 - 1075 (208 - 156)	1641 - 1227 (238 - 178)
1034 - 779 (150 - 113)	1241 - 931 (180 - 135)	1448 - 1083 (210 - 157)	1655 - 1241 (240 - 180)
1048 - 786 (152 - 114)	1225 - 936 (182 - 136)	1462 - 1089 (212 - 158)	1669 - 1248 (242 - 181)
1062 - 793 (154 - 115)	1269 - 952 (184 - 138)	1476 - 1103 (214 - 160)	1682 - 1262 (244 - 183)
1076 - 807 (156 - 117)	1282 - 965 (186 - 140)	1489 - 1117 (216 - 162)	1696 - 1269 (246 - 184)
1089 - 814 (158 - 118)	1296 - 972 (188 - 141)	1503 - 1124 (218 - 163)	1710 - 1282 (248 - 186)
1103 - 872 (160 - 120)	1310 - 979 (190 - 142)	1517 - 1138 (220 - 165)	1724 - 1289 (250 - 187)
1110 - 834 (161 - 121)	1324 - 993 (192 - 144)	1531 - 1145 (222 - 166)	

◆ **Interpreting Compression Readings**

1. If compression improves considerably, with the addition of oil, piston rings are faulty.
2. If compression does not improve with oil, valves are sticking or seating improperly.
3. If two adjacent cylinders indicate low compression pressures and squirting oil on each piston does not increase compression, the head gasket may be leaking between cylinders. Engine oil or coolant in cylinders could result from this condition.

◆ **TEST 12 – CHECK WIRE HARNESS**◆ **DISCUSSION**

Starting or running issues sometimes are caused by something as simple as a bad or intermittent electrical connection. Electrical connections can be very difficult to properly diagnose, especially when there is an intermittent connection.

NOTE:

Do not force meter leads into any housing or electrical connection, to do so may damage the housing or the electrical connection. Refer to PIB05-14-S for recommended wire removal, replacement and testing options.

◆ **PROCEDURE**

Verify that you have the proper electrical print (wiring, schematic or harness) for the unit that you are working with.

NOTE:

Disconnect the battery when checking continuity or you can get false readings.

Measure the resistance between the starting point and all ends of the individual circuits, and check for shorts to ground as well, while measuring the resistance move the connectors and wires looking for bad connections. Make sure that all ground connections are verified.

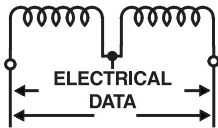
With circuits involving voltage, i.e. Wire 14, measure voltage on the circuit at all points looking for voltage drops (be sure to use the same ground reference point for all voltage measurements). Once again, while taking the readings move the wires and connectors and note any changes in the readings observed.

◆ **RESULTS**

Resistance readings should be within 0.2 Ohms of the resistance of your meter leads.

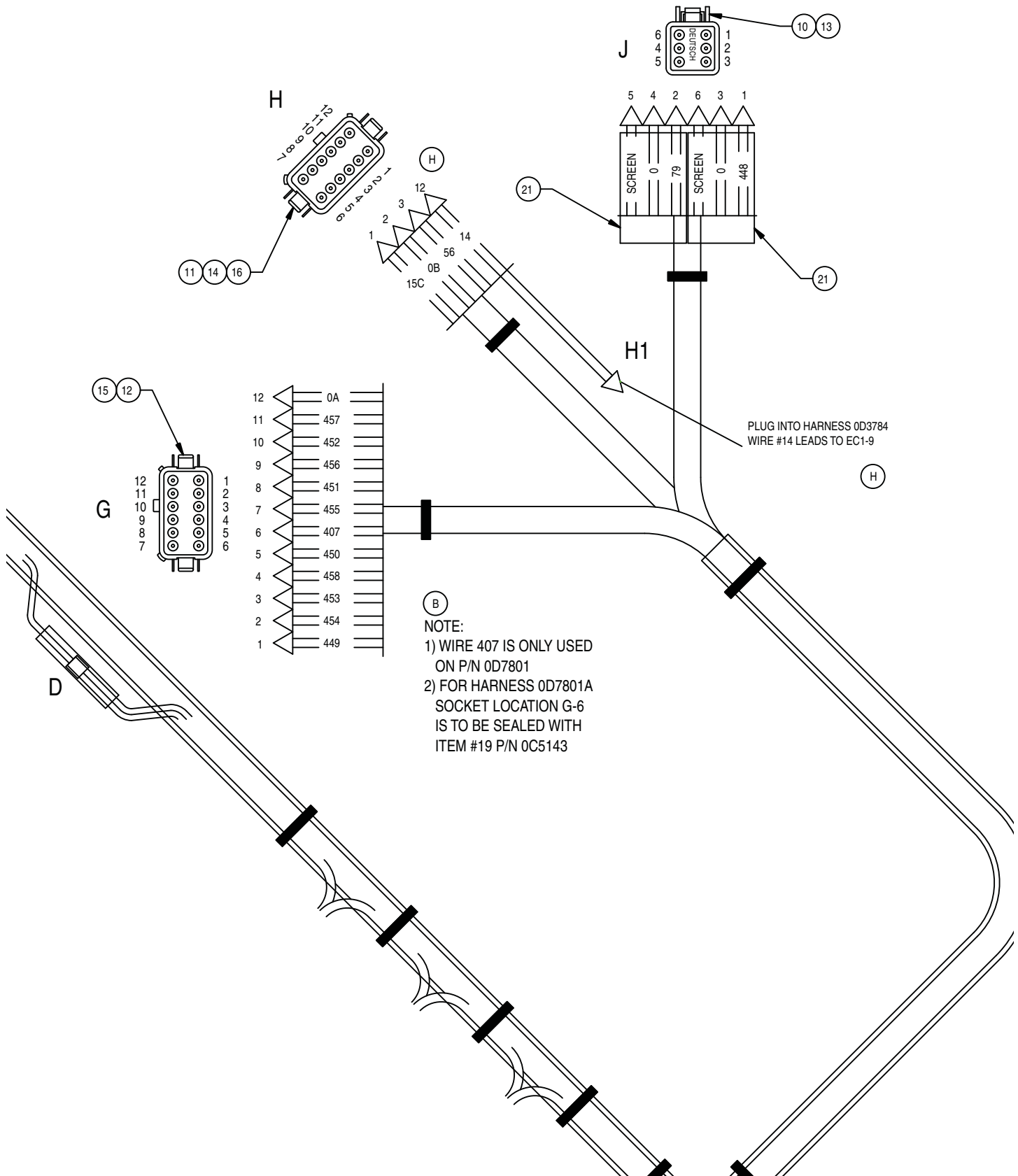
Voltage readings should be within +/- 5% of the source voltage reading.

If any readings are outside these tolerances, verify that all connections are clean and tight and retake the measurements. If the readings continue to remain outside the tolerances, repair or replace the bad connection(s) and/or wire(s).

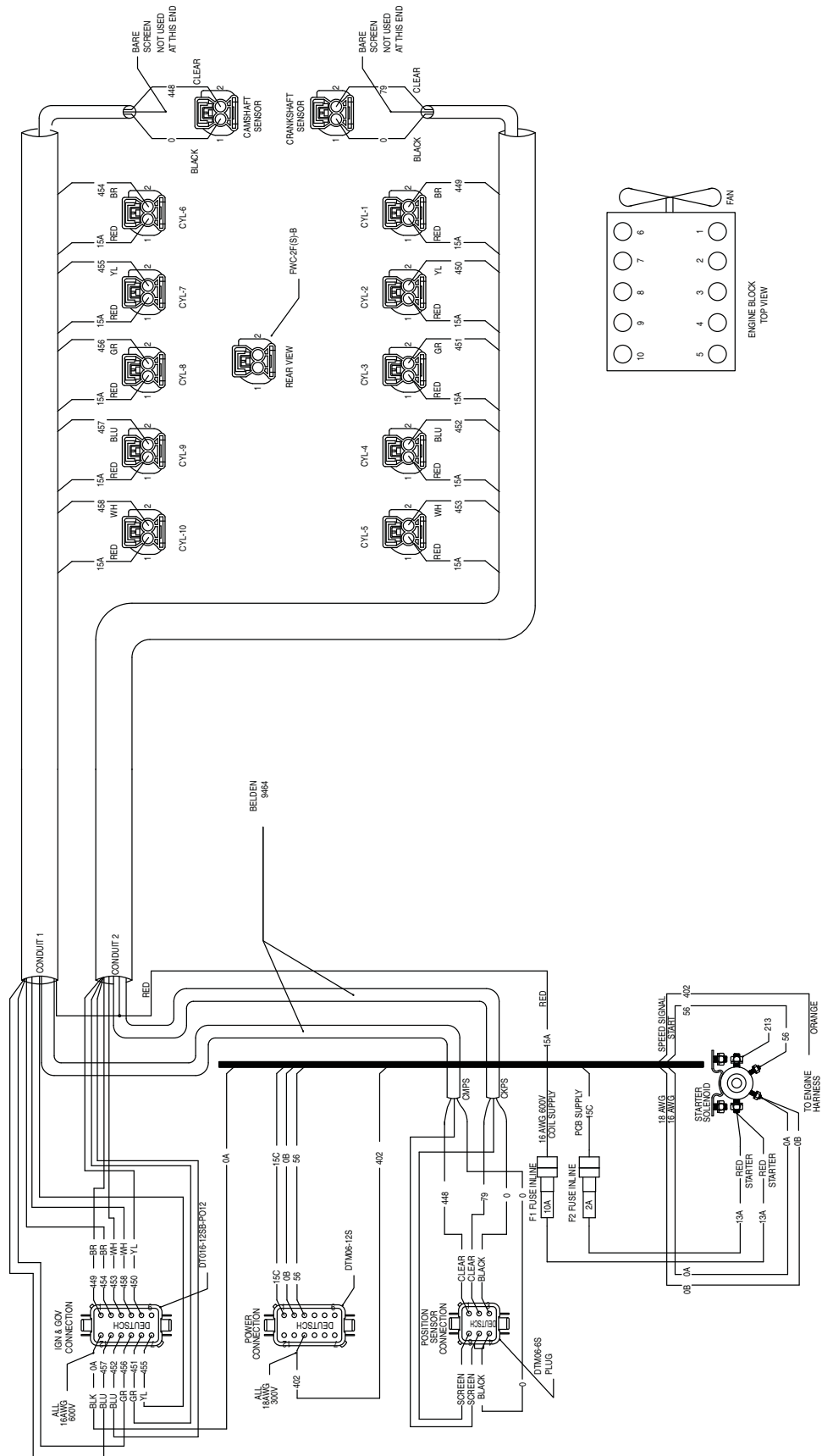
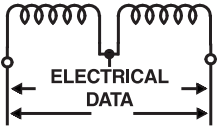


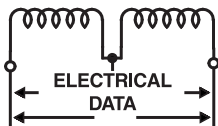
Electrical Data

C Panel & E Module 6.8L Ignition Harness - Drawing No. 0D7801\$-J



6.8L Ignition (Gear Box) Wiring Diagram - Drawing No. 0D8416-A





Electrical Data

4.6L, 5.4L & 6.8L H-panel Electrical Schematic - Drawing No. 0F3901-D Page 1

GD CONNECTOR

PIN	WIRE	TO	FUNCTION
1	0	GND	NOTE 1
4	14	RB1-7	NOTE 3
8	771	GA1-1	THROTTLE DRIVE LO
9	770	GA1-4	THROTTLE DRIVE HI
10	0	GND	NOTE 1
12	769	J1-33	THROTTLE PWM

AVR CONNECTOR

PIN	WIRE	TO	FUNCTION
1	1	FIELD	- FIELD
2	194	J2-31	+12VDC
3	6	PME	PME OUTPUT
4	4	R1	+ FIELD
5	4	FIELD	+ FIELD
6	403	J2-8	GATE TRIGGER B
7	404	J2-20	GATE TRIGGER A
10	406	J2-30	ZERO CROSSING I/P
11	405	J2-19	GROUND (ISO)
13	162	CB	PME OUTPUT (AFTER CB)

ICM - IGNITION MODULE CONNECTOR

PIN	WIRE	TO	FUNCTION
1	459	CYL9	IGNITION COIL DRIVE 9
2	460	CYL10	IGNITION COIL DRIVE 10
3	457	CYL7	IGNITION COIL DRIVE 7
4	451	CYL1	IGNITION COIL DRIVE 1
5	456	CYL6	IGNITION COIL DRIVE 6
6	452	CYL2	IGNITION COIL DRIVE 2
7	453	CYL3	IGNITION COIL DRIVE 3
8	454	CYL4	IGNITION COIL DRIVE 4
9	455	CYL5	IGNITION COIL DRIVE 5
10	SHLD	CUT	CAMSHAFT SENSOR DRAIN
11	56	RB1-10	STARTER RELAY OUT
12	14	RB1-7	NOTE 3
13	0	GND	NOTE 1
14	79A	CSS-2	CRANKSHAFT SENSOR +
15	458	CYL8	IGNITION COIL DRIVE 8
16	15C	F3	NOTE 7
17	0	GND	NOTE 1
18	15D	CYL1-CYL10	IGNITION COIL PWR
19	806	J1-27	IGNITION ALARM
20	SHLD	CUT	CRANKSHAFT SENSOR DRAIN
21	0A	CSS-1	CRANKSHAFT SENSOR -
22	0	CAM-1	CAMSHAFT SENSOR -
23	450	CAM-2	CAMSHAFT SENSOR +

ENGINE CONTROL MODULE CONNECTIONS

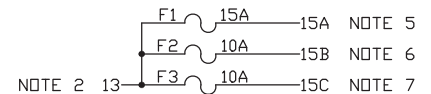
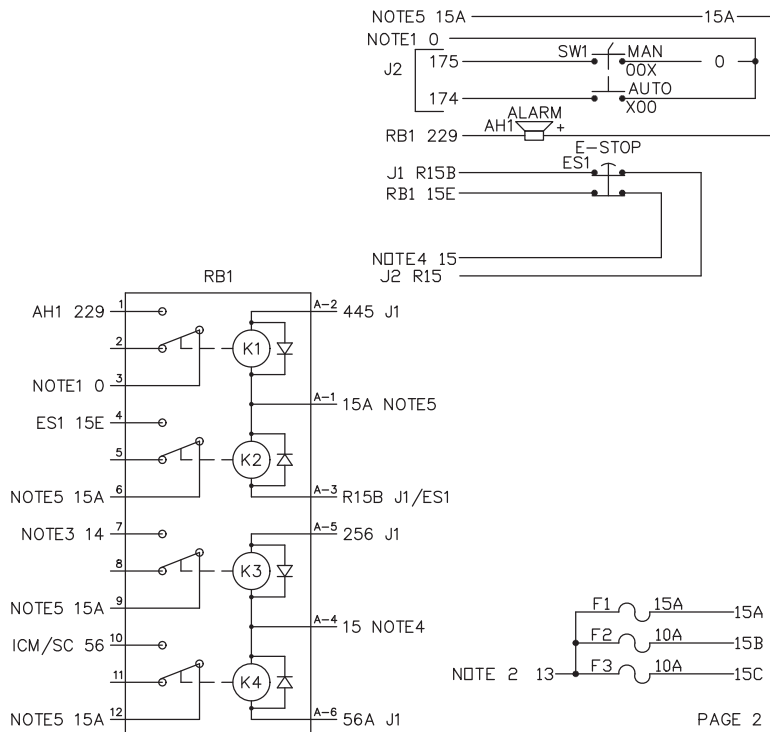
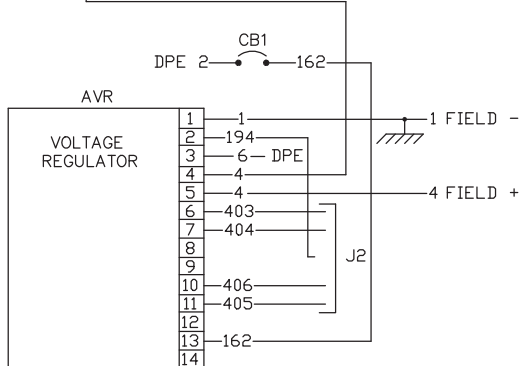
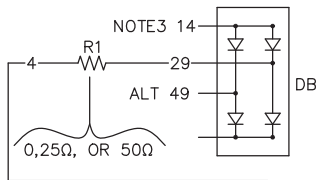
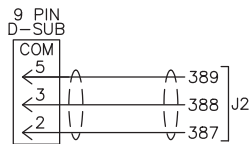
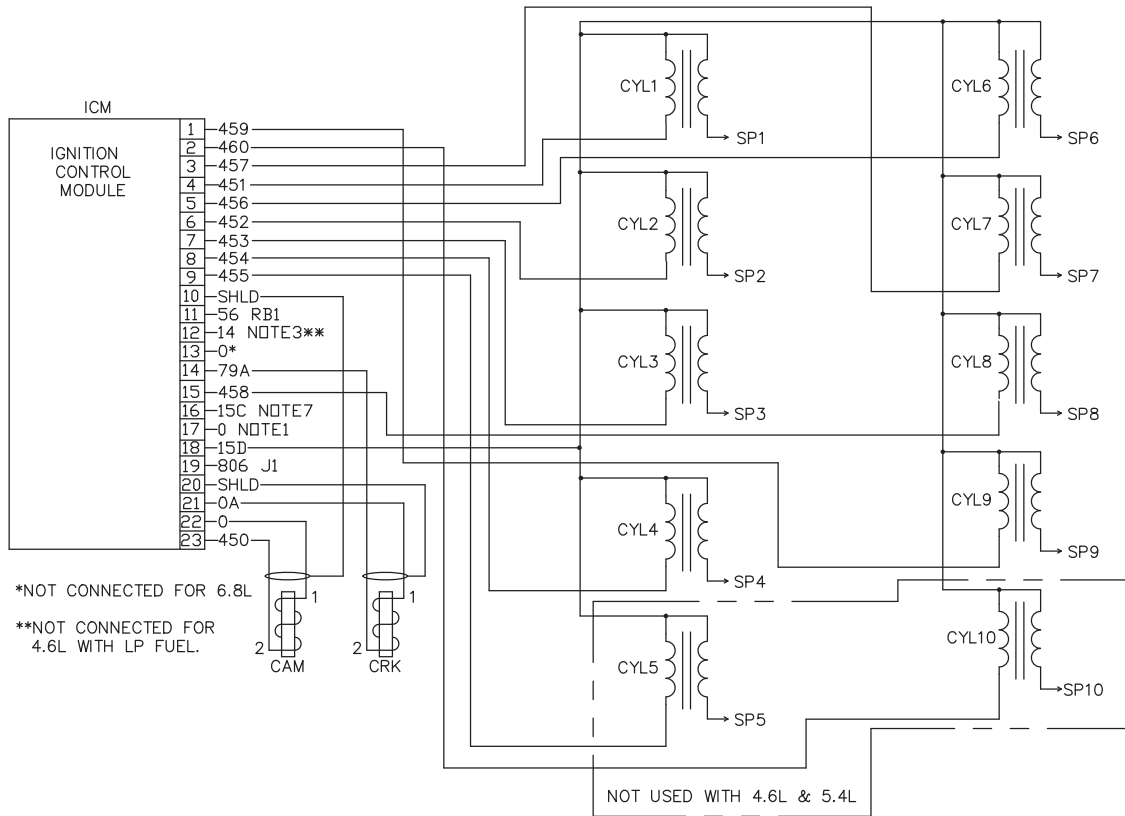
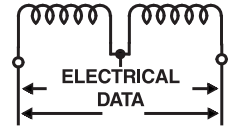
J1

PIN	WIRE	TO	FUNCTION
3	810	MOD-2	EXTERNAL MODEM GROUND
4	805	02	OXYGEN SENSOR RTN (OPTION)
5	804	02	OXYGEN SENSOR + (OPTION)
10	R15B	RB1A-3/ES1	OVERSPEED/WATCHDOG
11	256	RB1A-5	FUEL RELAY
12	0	GND	NOTE 1
14	811	MOD-3	DCD
15	68V	WTS-1	COOLANT TEMP +
16	803	BCH	BAT CHARGER CURRENT
17	766R	GA-2	THROTTLE POS RTN
18	766V	GA-1	THROTTLE POS +
19	69R	OPS-3	OIL PRESS RTN
20	69V	OPS-1	OIL PRESS +
21	808	AFS	AIR/FUEL SOLENOID (OPTION)
23	56A	RB1A-6	STARTER RELAY
24	0	MPU1-2	MPU1 SIGNAL (-)
25	79	MPU1-3	MPU1 SIGNAL (+)
26	812	MOD-4	ENABLE
27	806	ICM-19	IGNITION ALARM
29	573R	WLS-2	COOLANT LVL RTN
30	573V	WLS-1	COOLANT LVL +
31	68R	WTS-2	COOLANT TEMP RTN
32	809	MOD-1	+12
33	769	GD-12	THROTTLE PWM
34	445	RB1A-2	ALARM RELAY
35	15B	F2	NOTE 6

J2

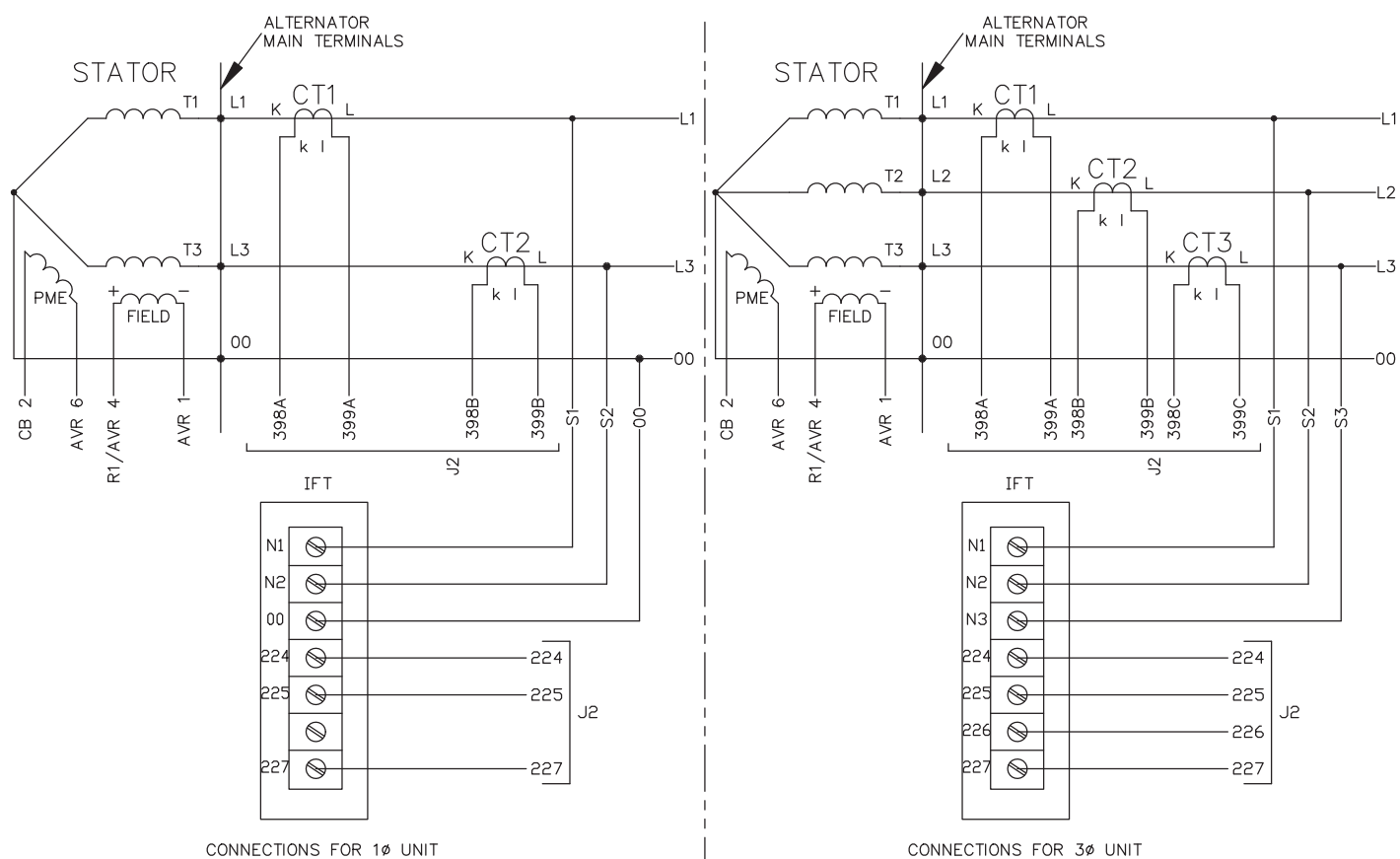
PIN	WIRE	TO	FUNCTION
1	391	CUST CON	RS485- (XFER SW)
2	388	COM-3	RS232 TX (GENLINK)
3	DI3	CUST CON	LINE POWER SIGNAL
4	183	CUST CON	REMOTE START
5	174	SW1	"AUTO" START
6	224	IFT	V SENSE GEN A PH
7	227	IFT	V SENSE RTN
8	403	AVR-6	AVR GATE TRIGGER B
* 9	399C	CT3	GEN C PH CURRENT -
* 10	398C	CT3	GEN C PH CURRENT +
11	399A	CT1	GEN A PH CURRENT -
12	398A	CT1	GEN A PH CURRENT +
13	390	CUST CON	RS485+ (XFER SW)
14	387	COM-2	RS232 RX (GENLINK)
15	601	LFP	LOW FUEL PRESSURE
16	R15	ES1	EMERGENCY STOP
* 17	226	IFT	V SENSE GEN C PH
19	405	AVR-11	AVR GROUND
20	404	AVR-7	AVR GATE TRIGGER A
24	SHLD	CUST CON	RS485 DRAIN (XFER SW)
25	389	COM-5	RS232 COM (GENLINK)
26	DI4	CUST CON	GEN POWER SIGNAL
27	505	BCH	BAT CHARGER FAIL
28	175	SW1	"MANUAL" START
29	225	IFT	V SENSE GEN B PH
30	406	AVR-10	AVR ZERO CROSSING I/P
31	194	AVR-2	AVR +12VDC
34	399B	CT2	GEN B PH CURRENT-
35	398B	CT2	GEN B PH CURRENT+

* - CONNECTIONS NOT USED IN 1Ø UNITS.

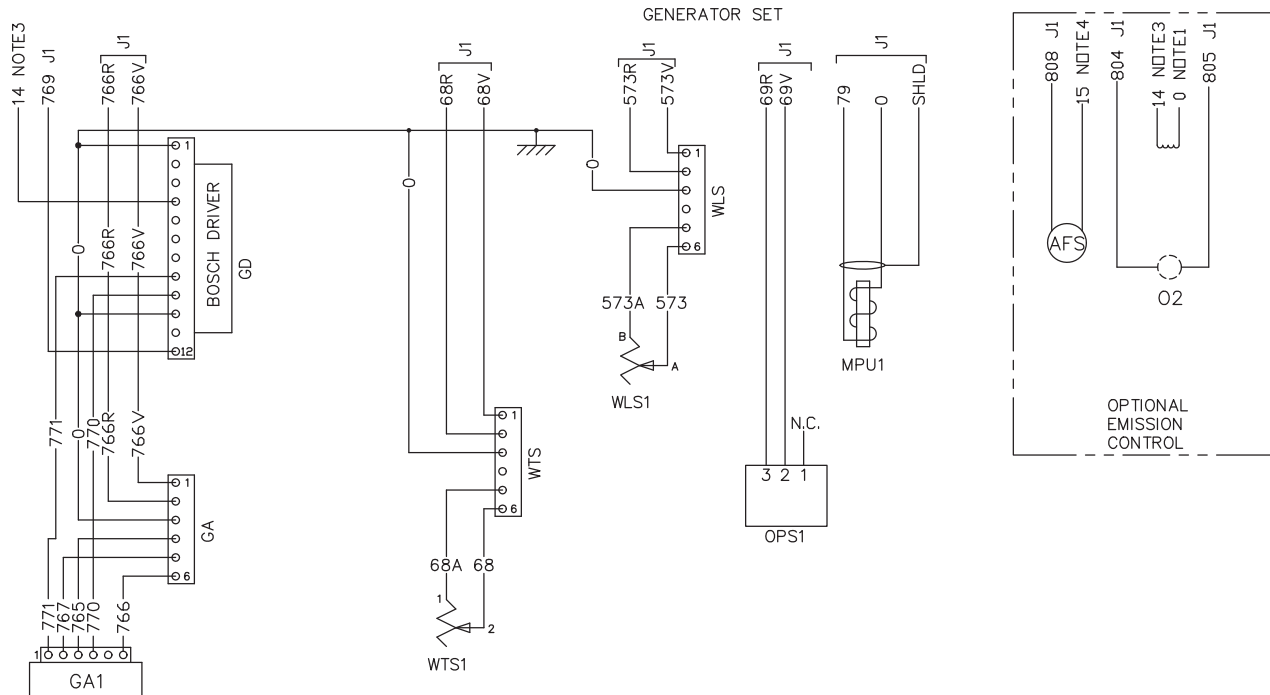


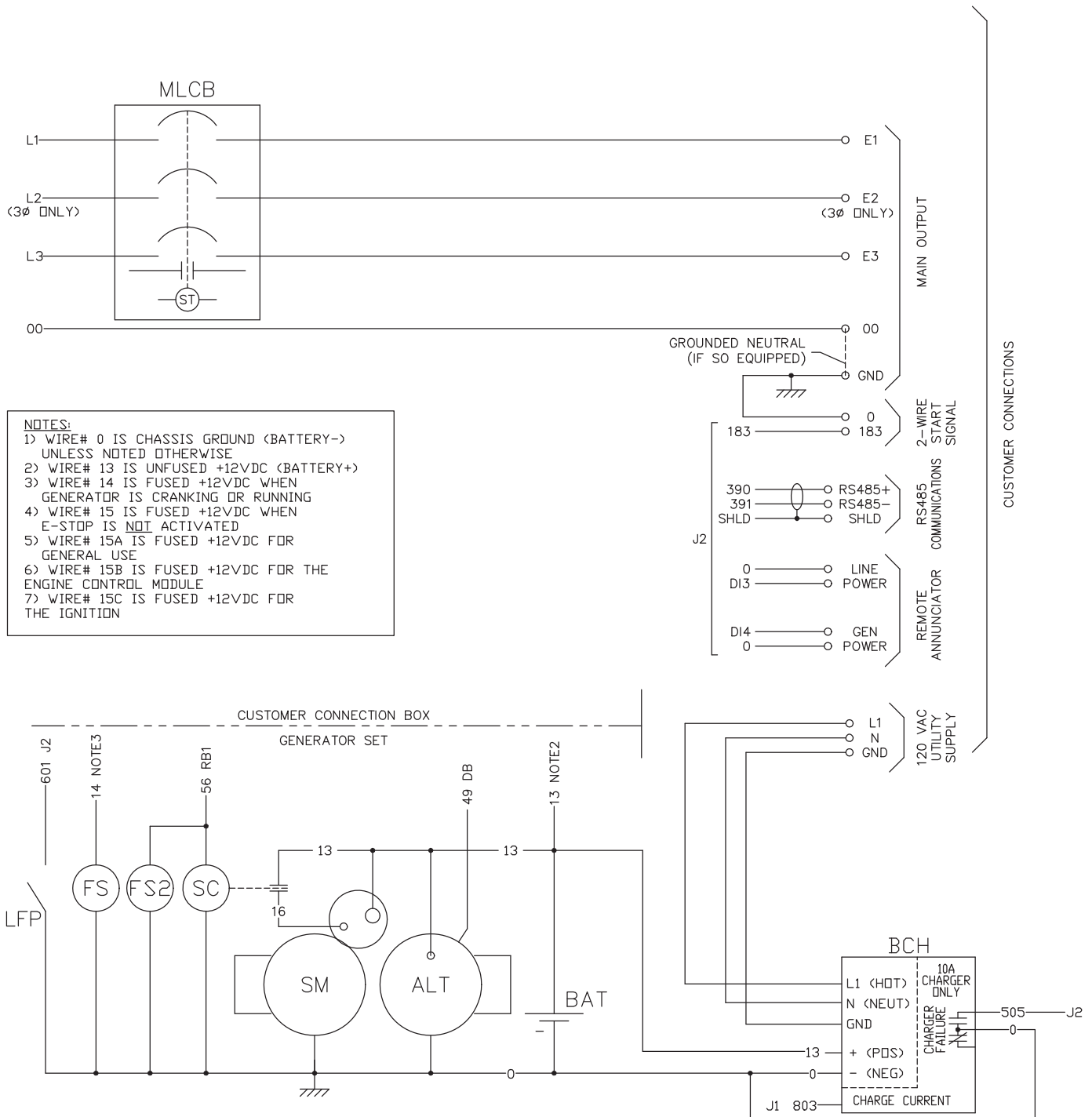
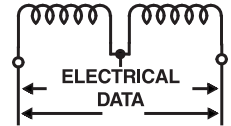
Electrical Data

4.6L, 5.4L & 6.8L H-panel Electrical Schematic - Drawing No. 0F3901-D Page 3



ALTERNATOR CONNECTION BOX



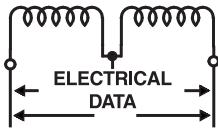


NOTES:

- 1) WIRE# 0 IS CHASSIS GROUND (BATTERY-)
UNLESS NOTED OTHERWISE
- 2) WIRE# 13 IS UNFUSED +12VDC (BATTERY+)
- 3) WIRE# 14 IS FUSED +12VDC WHEN
GENERATOR IS CRANKING OR RUNNING
- 4) WIRE# 15 IS FUSED +12VDC WHEN
E-STOP IS NOT ACTIVATED
- 5) WIRE# 15A IS FUSED +12VDC FOR
GENERAL USE
- 6) WIRE# 15B IS FUSED +12VDC FOR THE
ENGINE CONTROL MODULE
- 7) WIRE# 15C IS FUSED +12VDC FOR
THE IGNITION

LEGEND

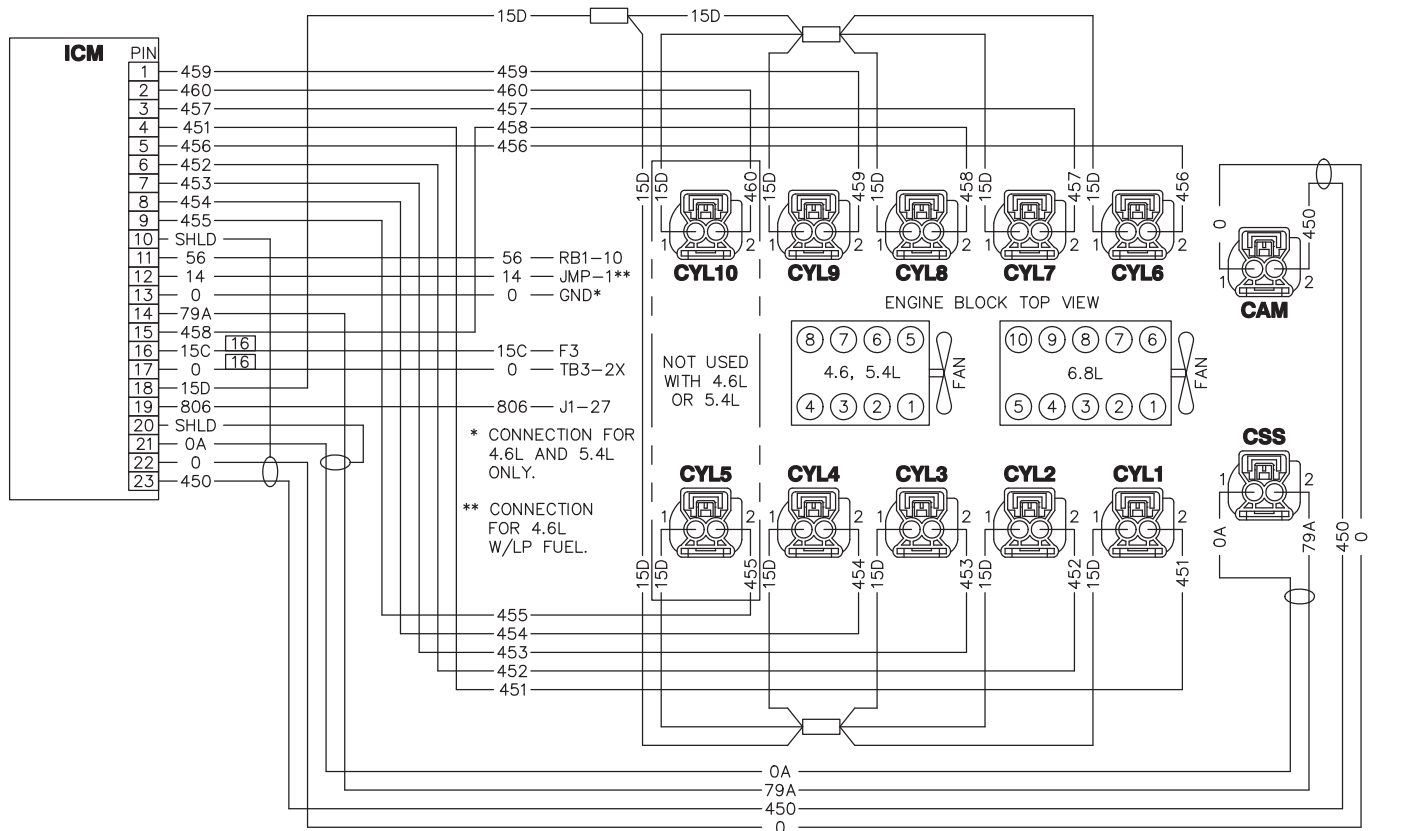
00 - NEUTRAL	CT - CURRENT TRANSFORMER	J - ENGINE CONTROLLER CONNECTIONS
AH1 - ALARM HORN	CUST CON - CUSTOMER CONNECTION	MLCB - MAIN LINE CIRCUIT BREAKER
ALT - CHARGE ALTERNATOR	CYL - CYLINDER IGNITION COIL	MPU - MAGNETIC PICK-UP
AVR - AUTOMATIC VOLTAGE REGULATOR	DPE - EXCITER	DPS - OIL PRESSURE SENDER
BAT - BATTERY (12VDC)	ES1 - E-STOP SWITCH	R - RESISTOR
BCH - BATTERY CHARGER	F - FUSE	RB - RELAY BOARD
CAM - CAMSHAFT SENSOR	FS - FUEL SOLENOID	SC - STARTER CONTACTOR
CB - CIRCUIT BREAKER	GA - GOVERNOR ACTUATOR	SHLD - SHIELD
COM - COMMUNICATION CONNECTOR	GD - GOVERNOR DRIVER	SM - STARTER MOTOR
CRK - CRANKSHAFT SENSOR	ICM - IGNITION CONTROL MODULE	SP - SPARK PLUG
	IFT - INTERFACE TRANSFORMER	ST - SHUNT TRIP
	ISD - ISOLATED (ELECTRICALLY)	SW1 - AUTO/MANUAL SELECT SWITCH
		VLS - WATER LEVEL SENSOR
		WTS - WATER TEMPERATURE SENDER



Electrical Data

4.6L, 5.4L & 6.8L H-panel Wiring Diagram - Drawing No. 0F2576-F Page 1

COMPONENTS LOCATED ON ENGINE



NOTE: ALL WIRES 18 AWG 300V UL LISTED UNLESS SHOWN OTHERWISE

AWG SIZE

S1, S2 & S3 ARE 600V UL LISTED

OPTION NOT AVAILABLE ON CPL UNITS

OTS1

GD

LOCATED IN CONTROL PANEL

ESC3

GA1

ESC1

ESC2

WTS1

WLS1

OPS1

MPU1

TB3-2Y 0

RB1-7 14

TB3-2Y 0

J1-33 769

GND 0

J1-17 766R

J1-18 766V

0

770

771

14

0

769

770

766

771

765

770

766

523V

523R

0

68V

68R

0

523V - J1-9

523R - J1-8

0

68V - J1-15

68R - J1-31

0

573V - J1-30

573R - J1-29

0

69V - J1-20

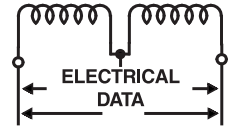
69R - J1-19

0

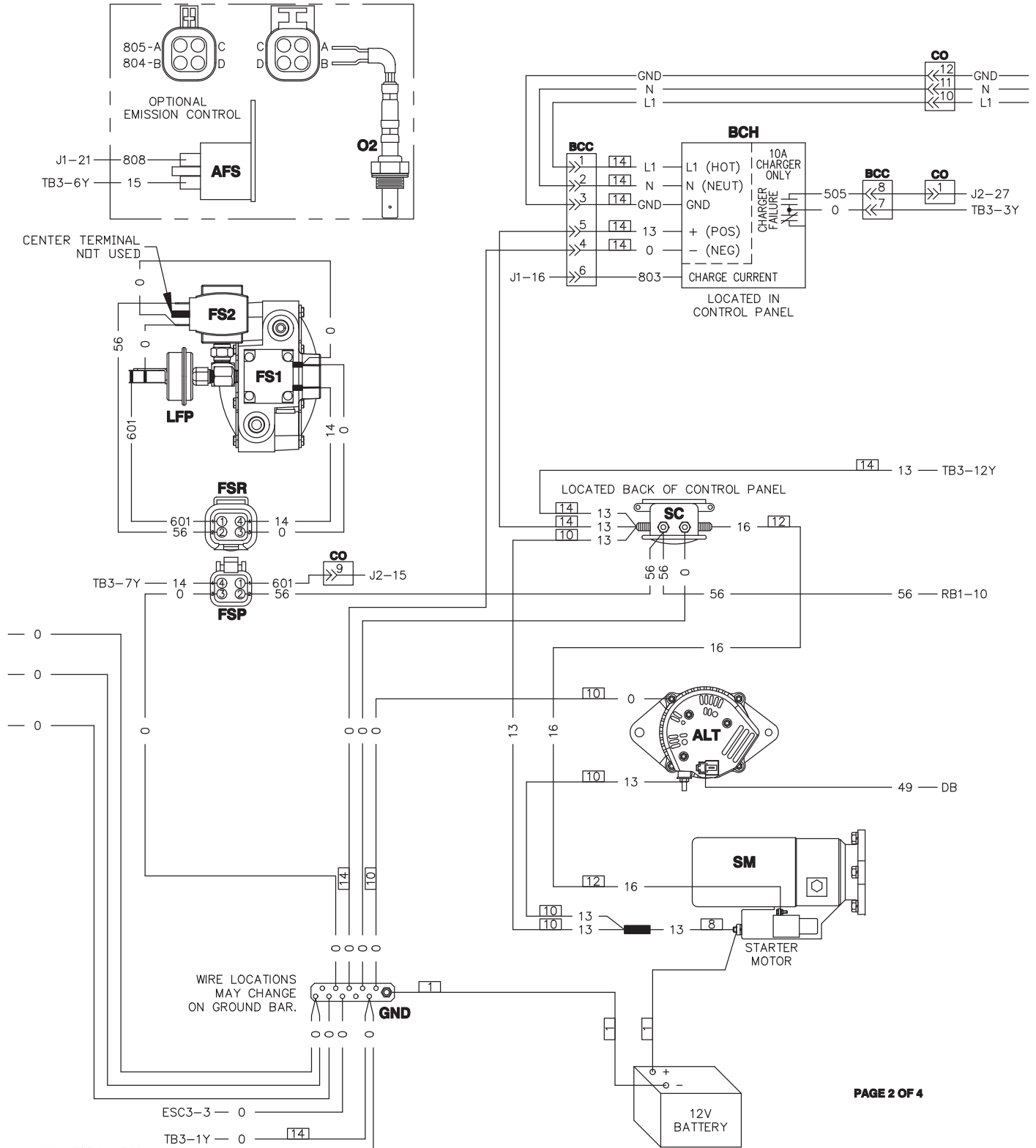
0 - J1-24

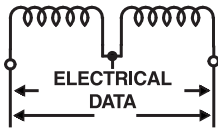
79 - J1-25

ECM CHASSIS



COMPONENTS LOCATED ON ENGINE

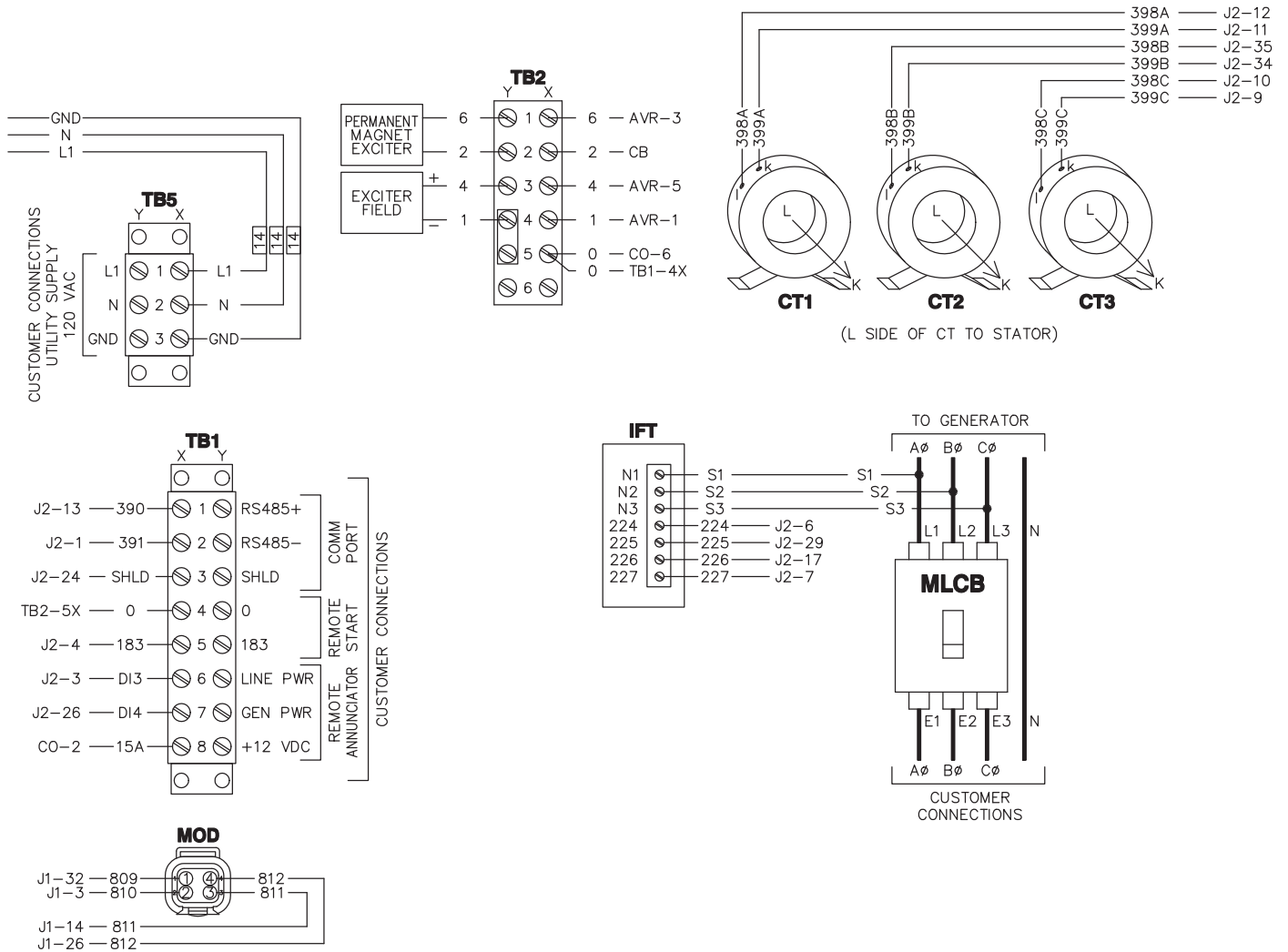




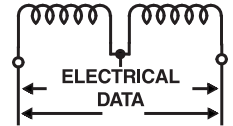
Electrical Data

4.6L, 5.4L & 6.8L H-panel Wiring Diagram - Drawing No. 0F2576-F Page 3

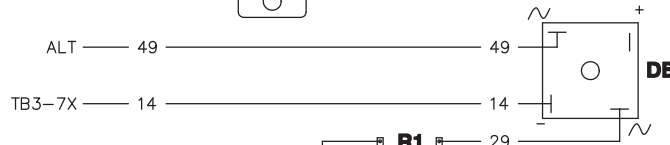
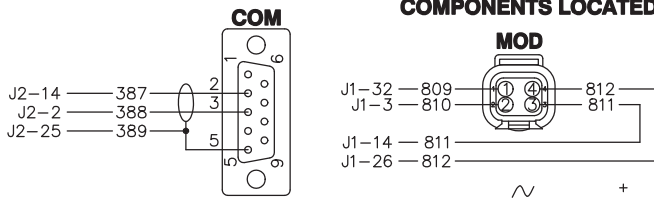
COMPONENTS LOCATED IN ALTERNATOR CONNECTION BOX



LEGEND	
AFS — AIR/FUEL SOLENOID	GD — GOVERNOR DRIVER
AH1 — ALARM HORN	GND — GROUND BAR CONNECTION
ALT — DC CHARGE ALTERNATOR	ICM — IGNITION CONTROL MODULE
AVR — AUTOMATIC VOLTAGE REGULATOR	IFT — INTERFACE TRANSFORMER
BCC — BATTERY CHARGER CONNECTOR	J_ — ELECTRONIC CONTROL MODULE CONNECTIONS
BCH — BATTERY CHARGER	JMP — JUMPER CONNECTION FOR NATURAL GAS
CAM — CAMSHAFT SENSOR	LFP — LOW FUEL PRESSURE SWITCH
CB — CIRCUIT BREAKER, DPE	MLCB — MAIN LINE CIRCUIT BREAKER
CO — CROSSOVER CONNECTOR	MOD — MODEM CONNECTOR
COM — COMMUNICATIONS PORT (9 PIN D-SUB)	MPU1 — MAG. PICK UP
CSS — CRANKSHAFT SENSOR	O2 — OXYGEN SENSOR
CT_ — CURRENT TRANSFORMER	OPS1 — OIL PRESSURE SENDER
CYL_ — CYLINDER IGNITION COIL	OTS1 — OIL TEMP. SENDER
DB — DIODE BRIDGE	R1 — RESISTOR
ECM — ELECTRONIC CONTROL MODULE (H-100)	RB1 — RELAY BOARD
ES1 — EMERGENCY STOP SWITCH	RB1A — RELAY BOARD CONNECTOR
ESC_ — ELECTRONIC SIGNAL CONDITIONER	SC — START CONTACTOR
F_ — FUSE	SM — STARTER MOTOR CONNECTIONS
FS_ — FUEL SOLENOID	SW1 — AUTO/MANUAL SELECT SWITCH
FSP — FUEL SOLENOID PLUG	SWC — OPERATOR SWITCH CONNECTOR
FSR — FUEL SOLENOID RECEPTACLE	TB_ — TERMINAL BLOCKS
GA1 — GOVERNOR ACTUATOR	WLS1 — COOLANT LEVEL SENDER
	WTS1 — COOLANT TEMP. SENDER

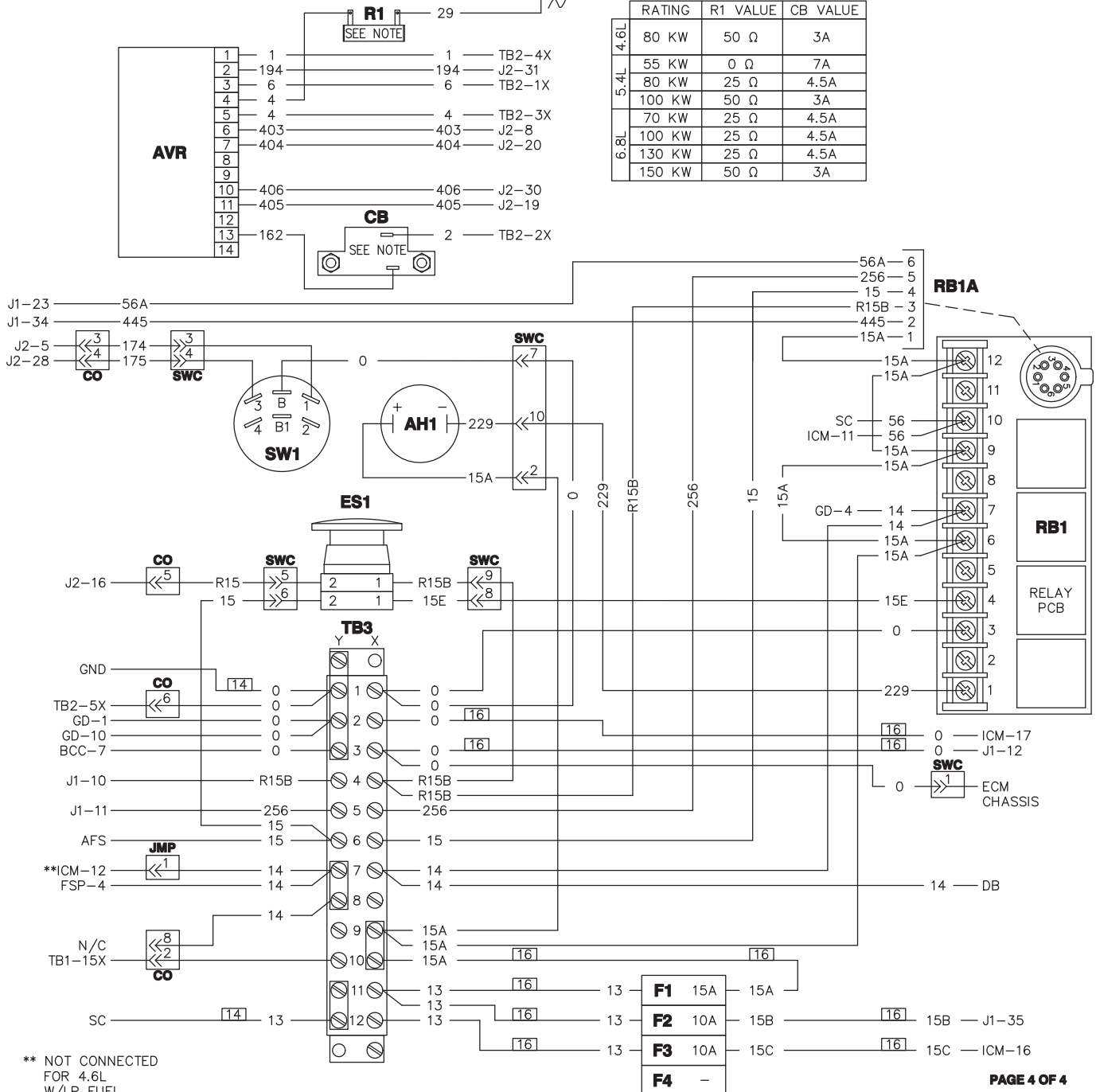


COMPONENTS LOCATED IN CONTROL PANEL

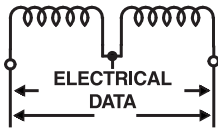


R1 AND CB VALUE BASED ON ENGINE AND KW RATING.

	RATING	R1 VALUE	CB VALUE
4.6L	80 KW	50 Ω	3A
	55 KW	0 Ω	7A
5.4L	80 KW	25 Ω	4.5A
	100 KW	50 Ω	3A
6.8L	70 KW	25 Ω	4.5A
	100 KW	25 Ω	4.5A
	130 KW	25 Ω	4.5A
	150 KW	50 Ω	3A



** NOT CONNECTED
FOR 4.6L
W/LP FUEL.



Electrical Data

5.4L & 6.8L PM-PC Electrical Schematic - Drawing No. 0F4494-E Page 1

BC1 - PCB6 CONNECTOR

PIN	WIRE#	TO	FUNCTION
1	425	J2-20	GEN VAB SENSE
2	426	J2-18	GEN VCA SENSE
3	424	J2-17	GEN VBC SENSE
4	PLUG		SPARE
5	0	J2-16	VDB1 GROUND
6	15	J2-15	NOTE 12
7	PLUG		SPARE
8	PLUG		SPARE
9	PLUG		SPARE
10	423	J2-13	BUS VAB SENSE
11	449	J2-19	BUS VCA SENSE
12	448	J2-14	BUS VBC SENSE

GD CONNECTOR

PIN	WIRE#	TO	FUNCTION
1	0	GND	NOTE 1
4	14	RB1-7	NOTE 3
8	771	GA1-1	THROTTLE DRIVE LO
9	770	GA1-4	THROTTLE DRIVE HI
10	0	GND	NOTE 1
12	769	J2-21	THROTTLE PWM

AVR CONNECTOR

PIN	WIRE#	TO	FUNCTION
1	1	FIELD	- FIELD
2	194	J2-34	+12VDC
3	6	PMG	PME OUTPUT
4	4	R1/FIELD	+ FIELD
5	4	R1/FIELD	+ FIELD
6	403	J2-32	GATE TRIGGER B
7	404	J2-31	GATE TRIGGER A
10	406	J2-33	ZERO CROSSING I/P
11	405	J2-35	GROUND (ISO)
13	162	CB1	PME OUTPUT (AFTER CB)

ICM - IGNITION MODULE CONNECTOR

PIN	WIRE#	TO	FUNCTION
1	459	CYL9	IGNITION COIL DRIVE 9
2	460	CYL10	IGNITION COIL DRIVE 10
3	457	CYL7	IGNITION COIL DRIVE 7
4	451	CYL1	IGNITION COIL DRIVE 1
5	456	CYL6	IGNITION COIL DRIVE 6
6	452	CYL2	IGNITION COIL DRIVE 2
7	453	CYL3	IGNITION COIL DRIVE 3
8	454	CYL4	IGNITION COIL DRIVE 4
9	455	CYL5	IGNITION COIL DRIVE 5
10	SHLD	CUT	CAMSHAFT SENSOR DRAIN
11	56	RB1-10	STARTER RELAY OUT
12	14	RB1-7	NOTE 3
13	0	GND	NOTE 1
14	79A	CRK-2	CRANKSHAFT SENSOR +
15	458	CYL8	IGNITION COIL DRIVE 8
16	15C	F3	NOTE 7
17	0	GND	NOTE 1
18	15D	CYL1-CYL10	IGNITION COIL PWR
19	806	J4-9	IGNITION ALARM
20	SHLD	CUT	CRANKSHAFT SENSOR DRAIN
21	0A	CRK-1	CRANKSHAFT SENSOR -
22	0	CAM-1	CAMSHAFT SENSOR -
23	450	CAM-2	CAMSHAFT SENSOR +

ENGINE CONTROL MODULE CONNECTIONS

J1

PIN	WIRE	TO	FUNCTION
1	0	MPU-2	MPU1 SIGNAL (-)
2	SHLD	CUT	RS232 DRAIN (GENLINK)
3	SHLD	CUT	RS232 DRAIN (DIAG)
11	0	OI	DISPLAY GROUND (ISO)
12	220C	OI	DISPLAY +24VDC
13	79	MPU-3	MPU1 SIGNAL (+)
14	SHLD	CUT	RS485 DRAIN (DISPLAY)
17	390	CUST CON	RS485+ (GENS)
18	391	CUST CON	RS485- (GENS)
19	SHLD	CUST CON	RS485 DRAIN (GENS)
20	392	OI	RS485+ (DISPLAY)
21	393	OI	RS485- (DISPLAY)
22	0	GND	NOTE 1
24	387	COM-2	RS232 RX (GENLINK)
25	389	COM-5	RS232 COM (GENLINK)
26	388	COM-3	RS232 TX (GENLINK)
27	387A	CUST CON	RS232 RX (DIAG)
28	389A	CUST CON	RS232 COM (DIAG)
29	388A	CUST CON	RS232 TX (DIAG)
31	800	CUST CON	MODEM (TIP)
32	801	CUST CON	MODEM (RING)
33	SHLD	MPU-1	MPU1 DRAIN
35	15B	F2	NOTE 6

J2

PIN	WIRE	TO	FUNCTION
7	398A	CT1	GEN AØ CURRENT +
8	399A	CT1	GEN AØ CURRENT -
9	398B	CT2	GEN BØ CURRENT +
10	399B	CT2	GEN BØ CURRENT -
11	398C	CT3	GEN CØ CURRENT +
12	399C	CT3	GEN CØ CURRENT -
13	423	BC1-10	BUS VAB SENSE
14	448	BC1-12	BUS VBC SENSE
15	15	BC1-6	NOTE 12
16	0	BC1-5	VOLT DIV GROUND
17	424	BC1-3	GEN VBC SENSE
18	426	BC1-2	GEN VCA SENSE
19	449	BC1-11	BUS VCA SENSE
20	425	BC1-1	GEN VAB SENSE
21	769	GD-12	THROTTLE PWM
31	404	AVR-7	AVR GATE TRIGGER A
32	403	AVR-6	AVR GATE TRIGGER B
33	406	AVR-10	AVR ZERO CROSSING I/P
34	194	AVR-2	AVR +12VDC
35	405	AVR-11	AVR GROUND

J3

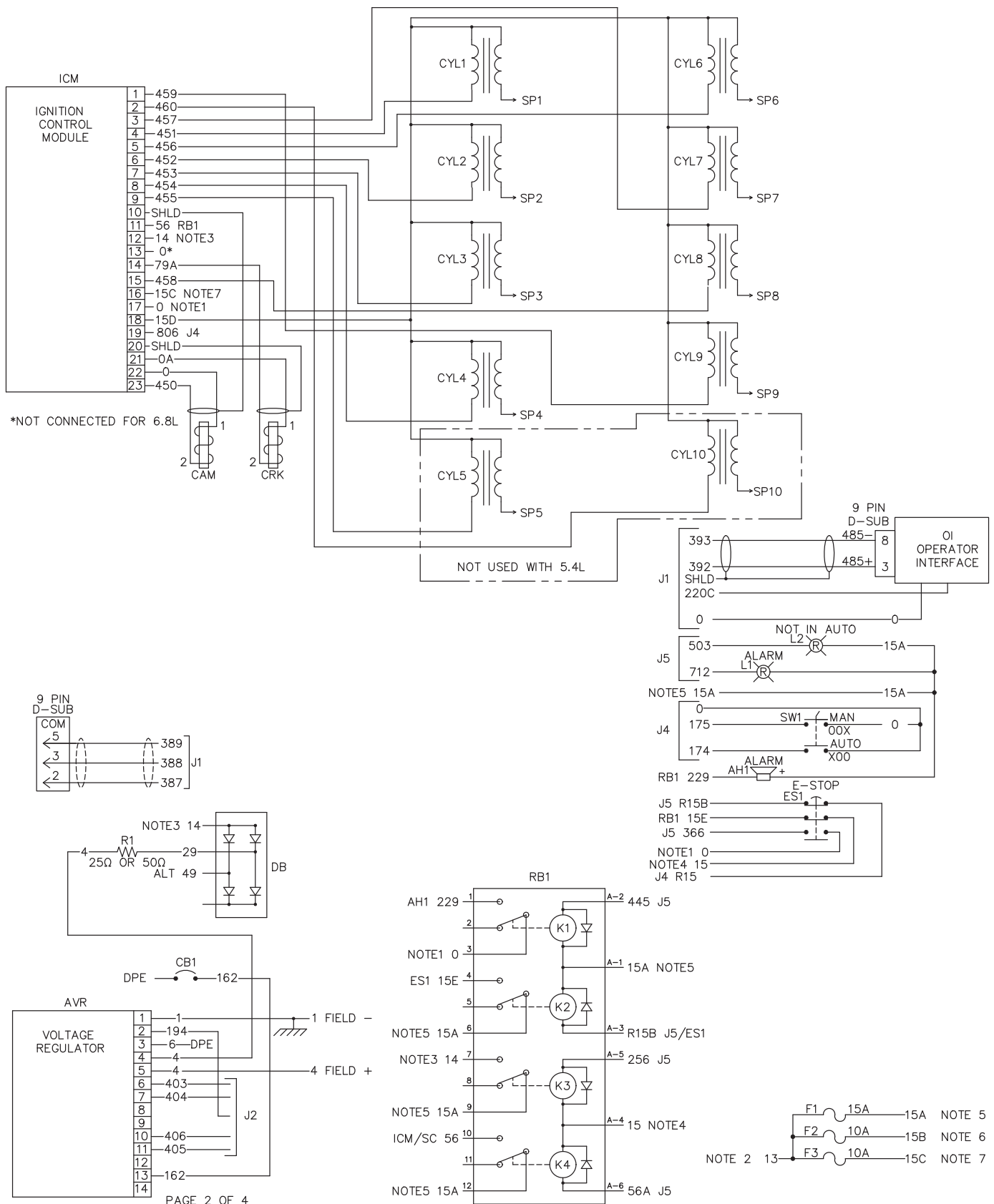
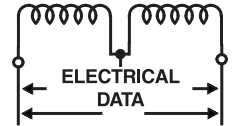
PIN	WIRE	TO	FUNCTION
3	68V	WTS-1	COOLANT TEMP +
4	68R	WTS-2	COOLANT TEMP RTN
5	69V	OPS-1	OIL PRESS +
6	69R	OPS-3	OIL PRESS RTN
7	766V	GA-1	THROTTLE POS +
8	766R	GA-2	THROTTLE POS RTN
19	573V	WLS-1	COOLANT LVL +
20	573R	WLS-2	COOLANT LVL RTN

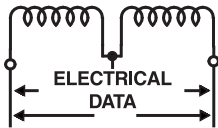
J4

PIN	WIRE	TO	FUNCTION
1	174	SW1	"AUTO" START
2	175	SW1	"MANUAL" START
4	R15	ES1	EMERGENCY STOP
5	183	CUST CON	REMOTE START
6	505	BCH	BAT CHARGER FAIL
8	601	LFP	LOW FUEL PRESSURE
9	806	ICM-19	IGNITION ALARM
13	371	CON1	CONTACTOR STATUS
16	418	MLCB	MLCB STATUS
27	D13	CUST CON	LINE PWR INPUT SIGNAL
28	D14	CUST CON	GEN PWR INPUT SIGNAL
31	COM	MLCB/CON	DI RETURN
32	0	SW1	DI RETURN

J5

PIN	WIRE	TO	FUNCTION
1	56A	RB1A-6	STARTER RELAY
2	256	RB1A-5	FUEL RELAY
3	445	RB1A-2	ALARM RELAY
7	503	L2	"AUTO" LED
9	712	L1	"ALARM" LED
12	366	TG-14	TRIP CONTACTOR
14	23	CG-14	CLOSE CONTACTOR
27	263A	ST-14	MLCB SHUNT TRIP
28	R15B	RB1A-3/ES1	OVERSPEED/WATCHDOG
33	0	GND	NOTE 1



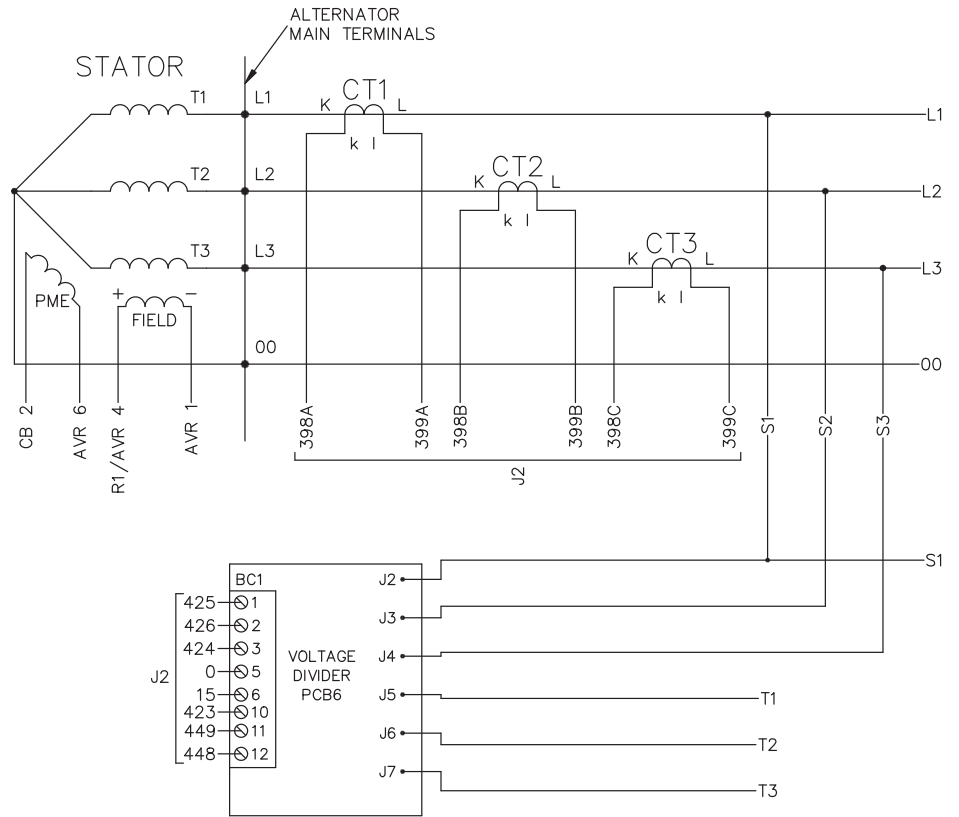


Electrical Data

5.4L & 6.8L PM-PC Electrical Schematic - Drawing No. 0F4494-E Page 3

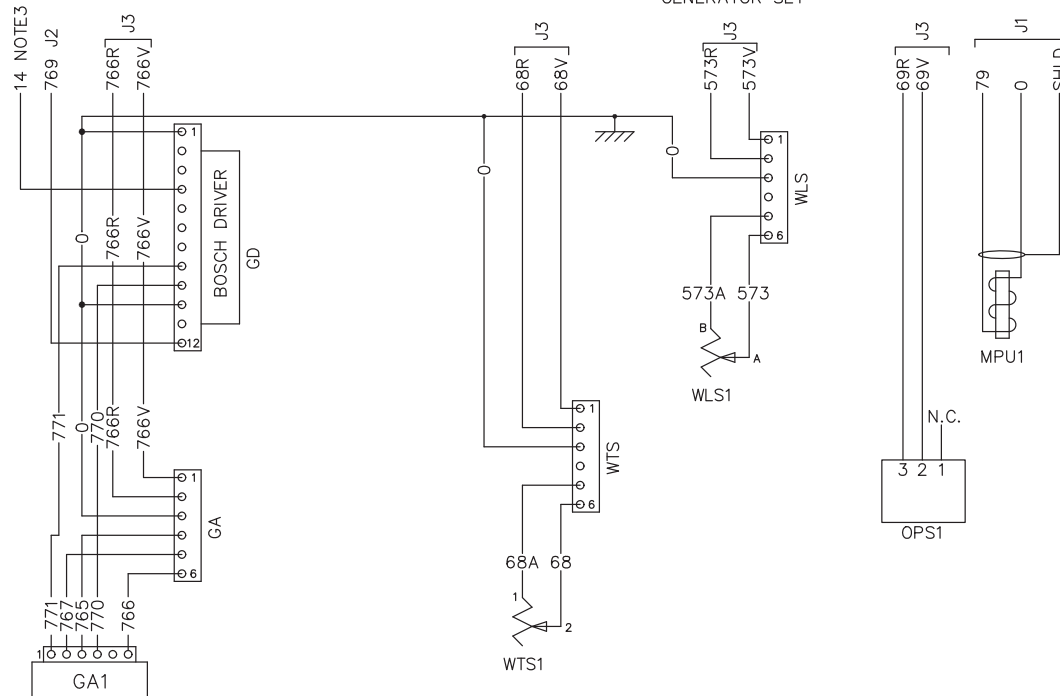
NOTES:

- 1) WIRE# 0 IS CHASSIS GROUND (BATTERY-) UNLESS NOTED OTHERWISE
- 2) WIRE# 13 IS UNFUSED +12VDC (BATTERY+)
- 3) WIRE# 14 IS FUSED +12VDC WHEN GENERATOR IS CRANKING OR RUNNING
- 4) WIRE# 15 IS FUSED +12VDC WHEN E-STOP IS NOT ACTIVATED
- 5) WIRE# 15A IS FUSED +12VDC FOR GENERAL USE
- 6) WIRE# 15B IS FUSED +12VDC FOR THE ENGINE CONTROL MODULE
- 7) WIRE# 15C IS FUSED +12VDC FOR THE IGNITION

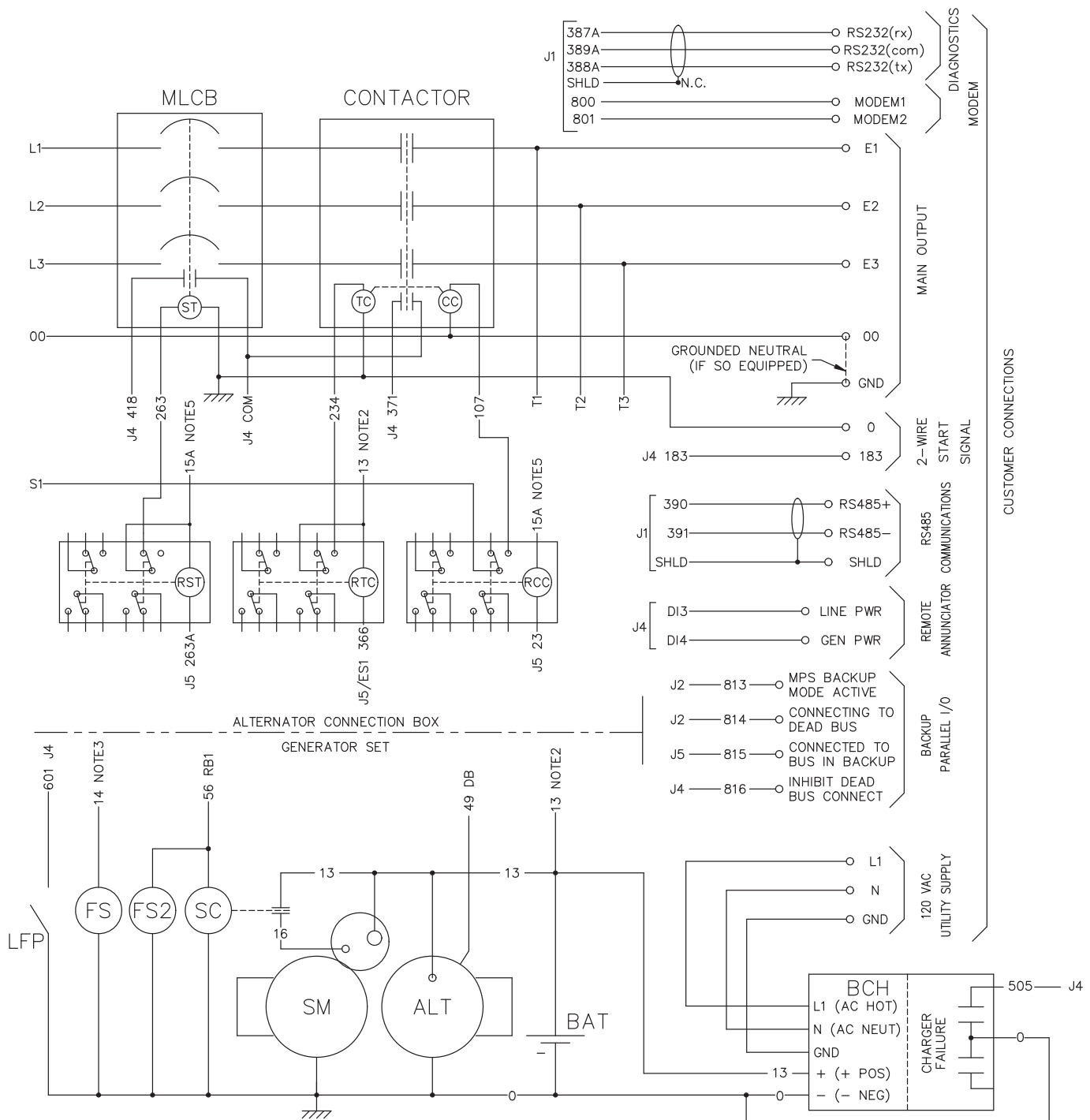
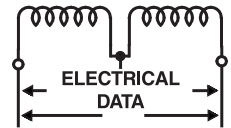


ALTERNATOR CONNECTION BOX

GENERATOR SET



5.4L & 6.8L PM-PC Electrical Schematic - Drawing No. 0F4494-E Page 4



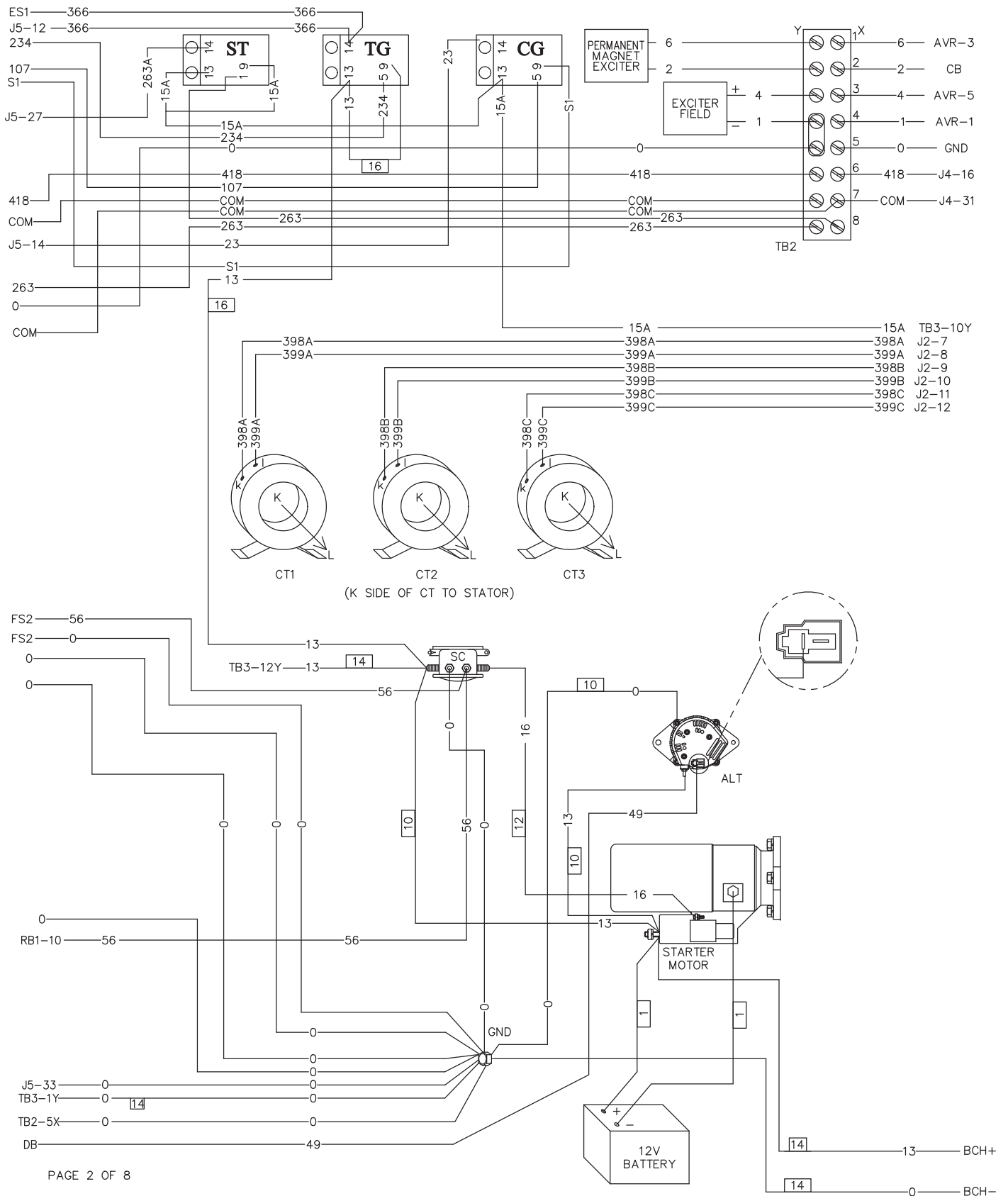
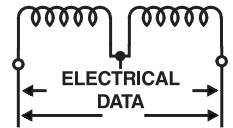
LEGEND

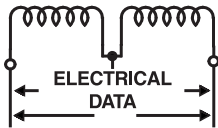
00 - NEUTRAL
 AH1 - ALARM HORN
 ALT - CHARGE ALTERNATOR
 AVR - AUTOMATIC VOLTAGE REGULATOR
 BAT - BATTERY (12VDC)
 BC1 - VOLTAGE DIVIDER (PCB6)
 BCH - BATTERY CHARGER
 CAM - CAMSHAFT SENSOR
 CB - CIRCUIT BREAKER
 COM - COMMUNICATION CONNECTOR

CRK	- CRANKSHAFT SENSOR
CT	- CURRENT TRANSFORMER
CUST CON	- CUSTOMER CONNECTION
CYL	- CYLINDER IGNITION COIL
DPE	- EXCITER
ES1	- E-STOP SWITCH
F	- FUSE
FS	- FUEL SOLENOID
GA	- GOVERNOR ACTUATOR
GD	- GOVERNOR DRIVER
ICM	- IGNITION CONTROL MODULE
ISO	- ISOLATED (ELECTRICALLY)

J_	-	ENGINE CONTROLLER CONNECTION
L_	-	INDICATOR LAMP
MLCB	-	MAIN LINE CIRCUIT BREAKER
MPU	-	MAGNETIC PICK-UP
OI	-	OPERATOR INTERFACE
OPS_	-	OIL PRESSURE SENDER
R_	-	RESISTOR
RB_	-	RELAY BOARD
SC	-	STARTER CONTACTOR
SHLD	-	SHIELD
SM	-	STARTER MOTOR
SP	-	SPARK PLUG

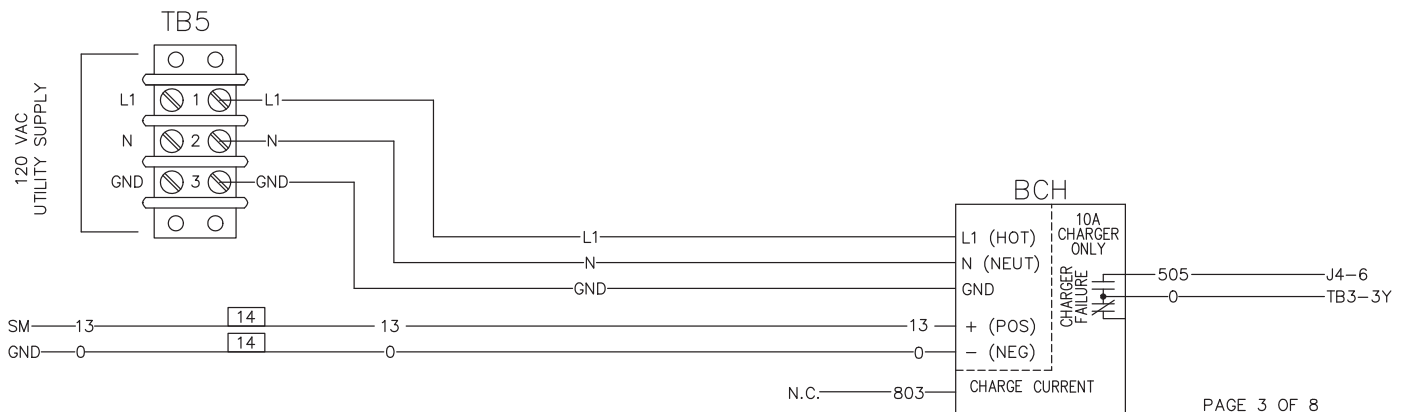
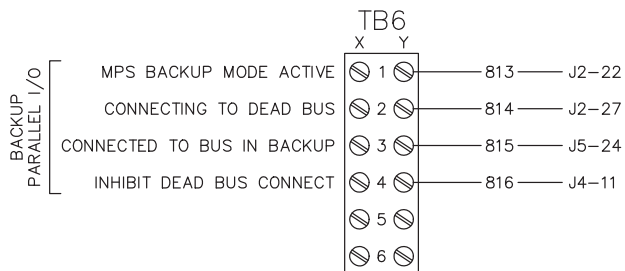
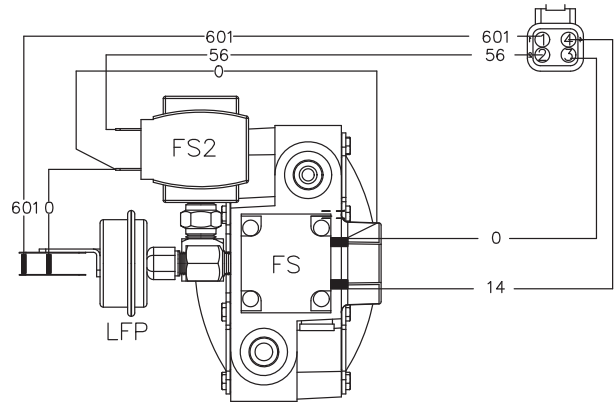
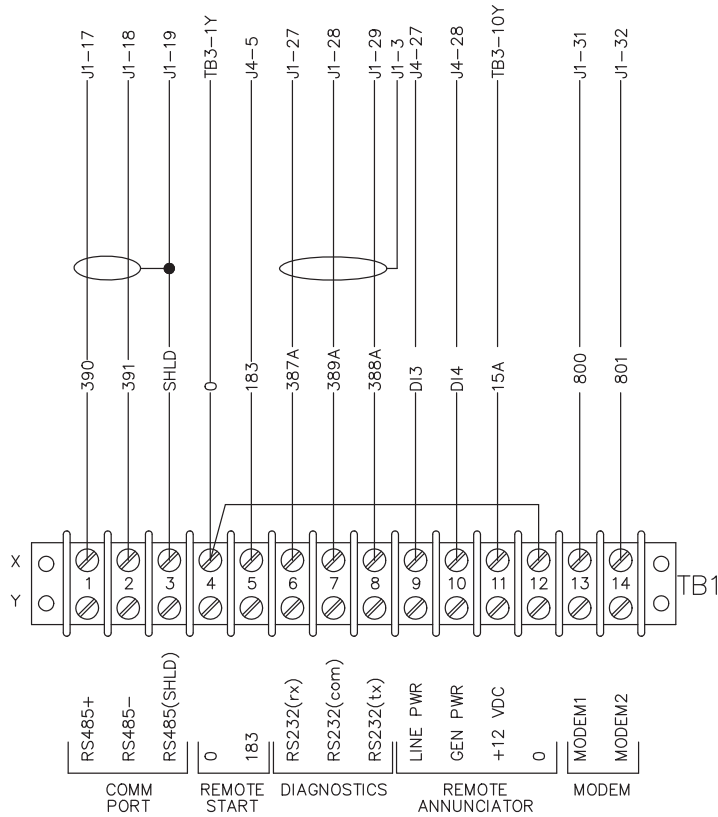
ST - SHUNT TRIP
SW1 - AUTO/MANUAL SELECT SWITCH
WLS_ - WATER LEVEL SENSOR
WTS_ - WATER TEMPERATURE SENDER

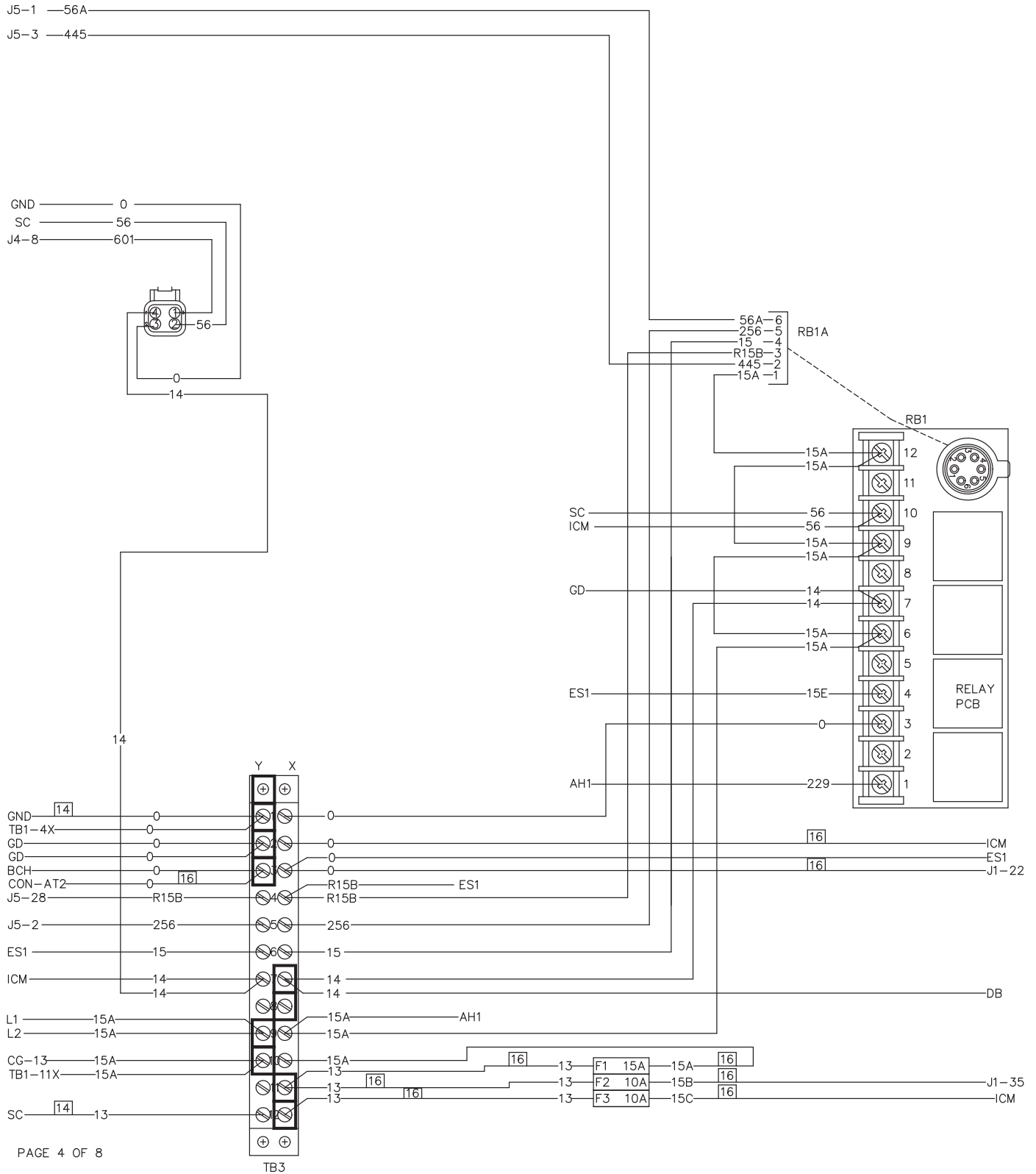
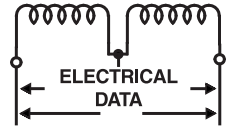


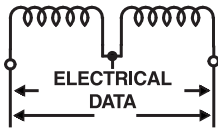


Electrical Data

5.4L & 6.8L PM-PC Wiring Diagram - Drawing No. 0F2837-F Page 3

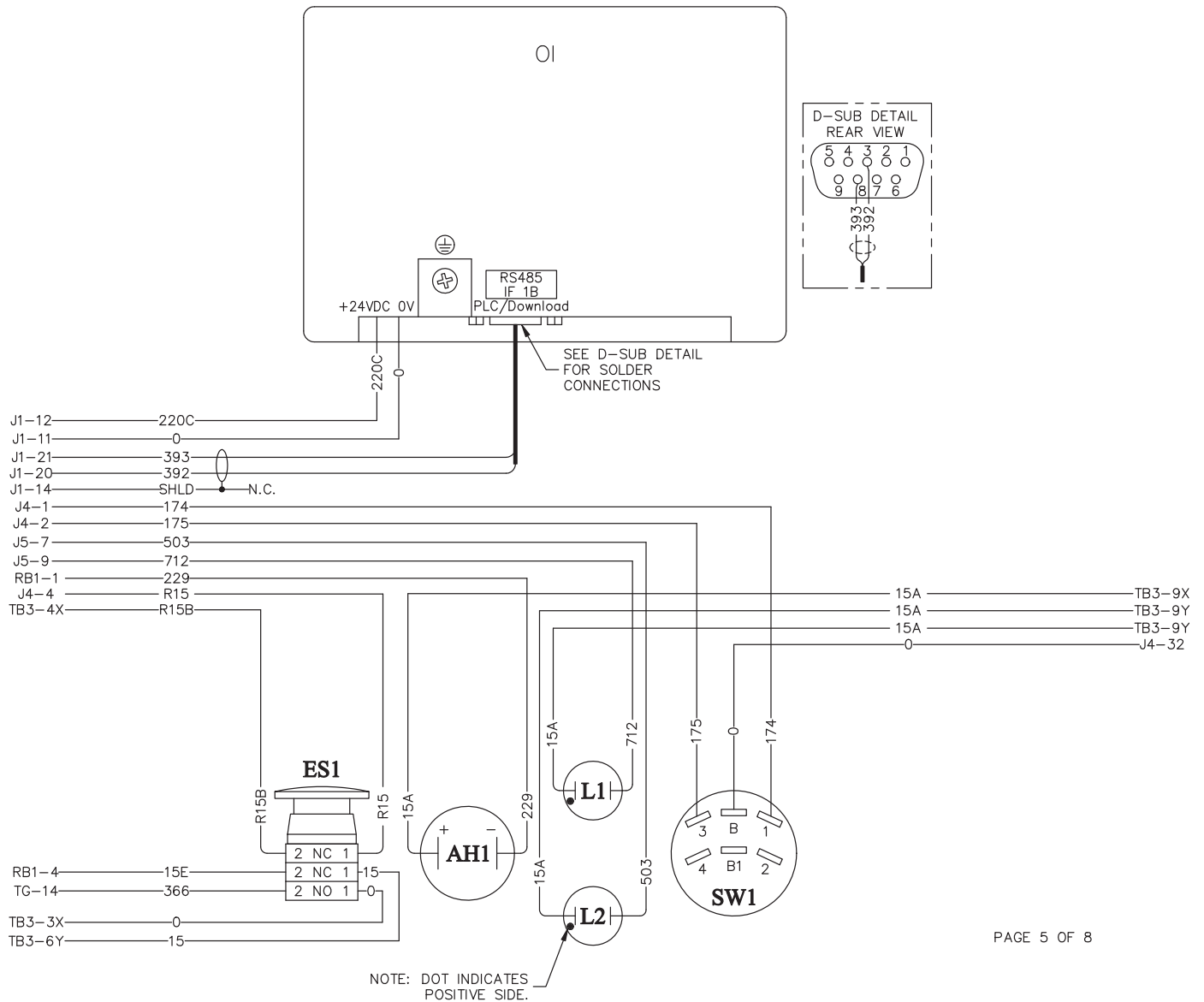
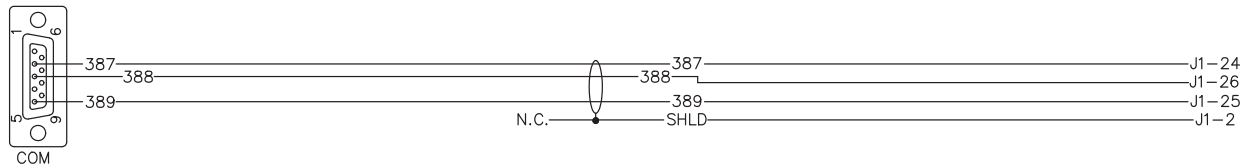


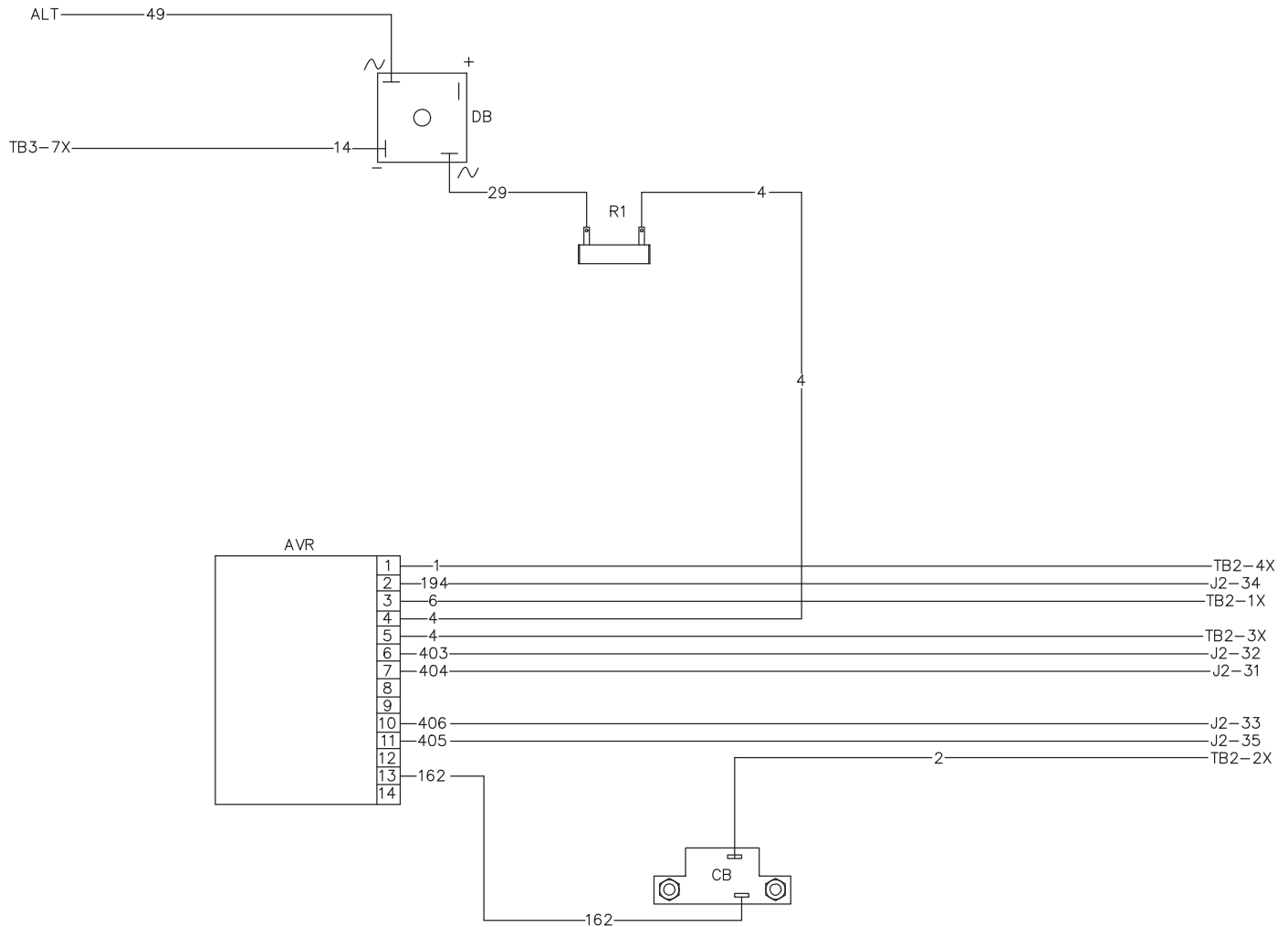
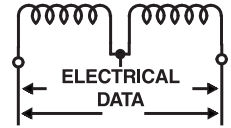




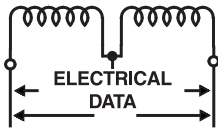
Electrical Data

5.4L & 6.8L PM-PC Wiring Diagram - Drawing No. 0F2837-F Page 5



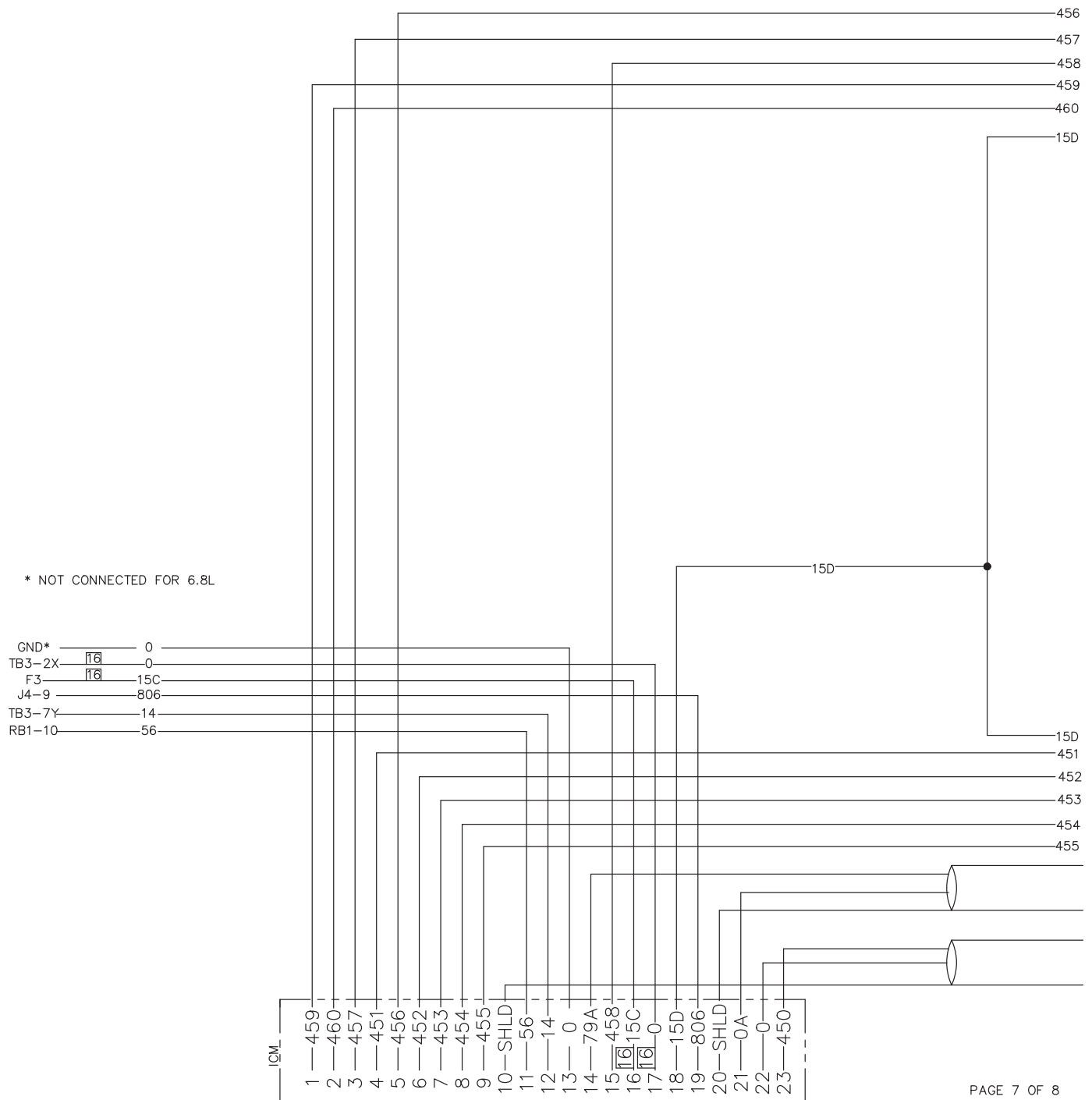


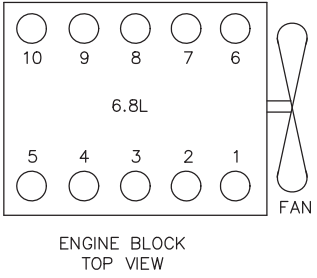
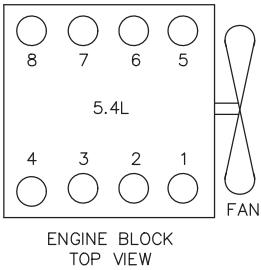
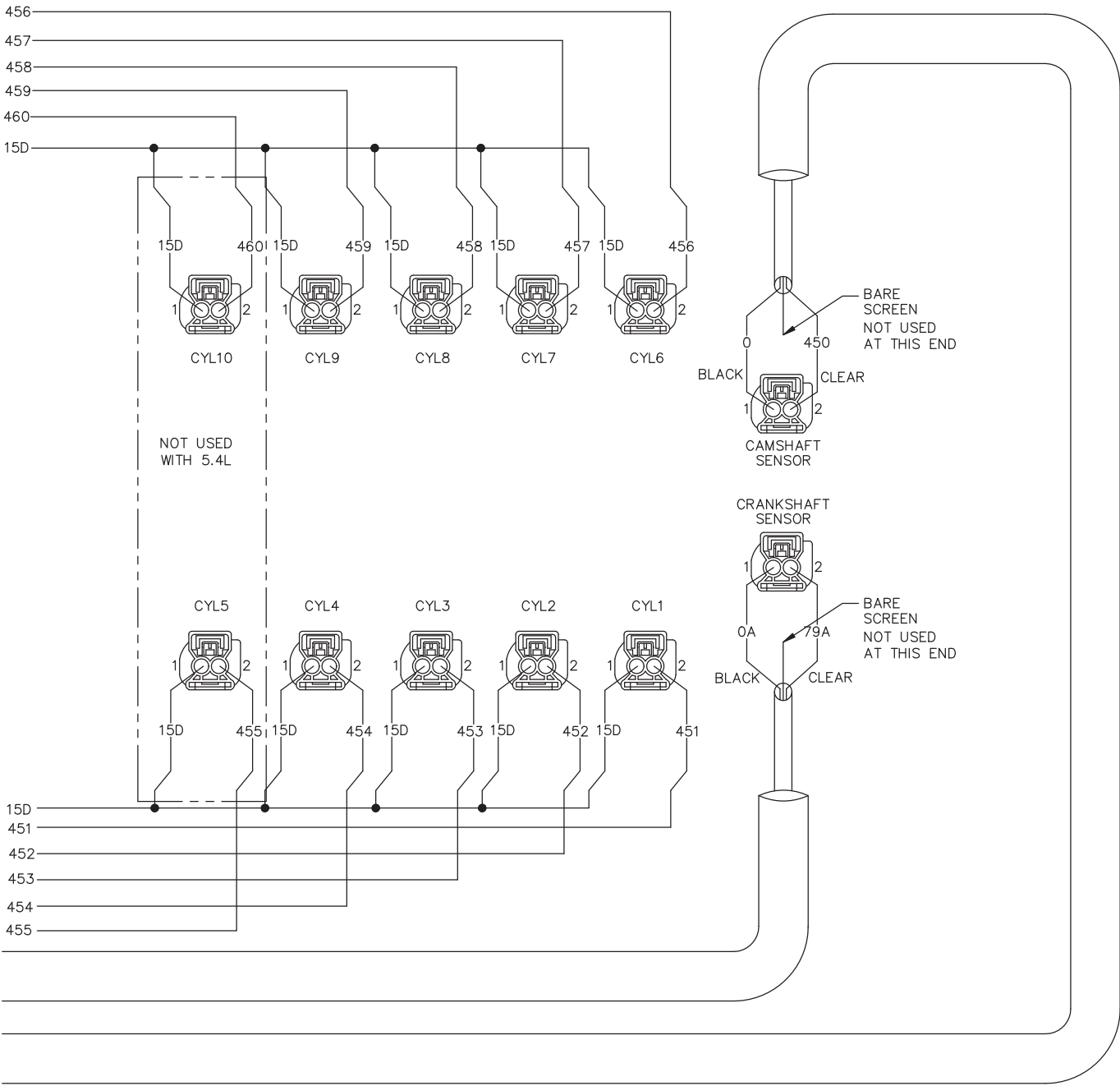
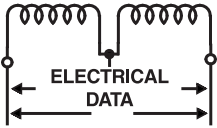
LEGEND	
AH1	- ALARM HORN
ALT	- DC CHARGE ALTERNATOR
AVR	- AUTOMATIC VOLTAGE REGULATOR
BCH	- BATTERY CHARGER
BC1	- VOLTAGE DIVIDER CONNECTOR (BLACK 12 PIN)
CG	- CLOSE CONTACTOR RELAY BASE
CB	- CIRCUIT BREAKER, DPE
COM	- COMMUNICATIONS PORT (9 PIN D-SUB)
CON	- CONTACTOR CONNECTIONS
CT1	- CT AØ
CT2	- CT BØ
CT3	- CT CØ
DB	- DIODE BRIDGE
ES1	- EMERGENCY STOP SWITCH
ESC	- ELECTRONIC SIGNAL CONDITIONER
F	- FUSE
FS	- FUEL SOLENOID
FS2	- AUX. FUEL SOLENOID
GA	- GAS ACTUATOR
GD	- GAS DRIVER (14 PIN)
GND	- GROUND STUD CONNECTION
ICM	- IGNITION CONTROL MODULE
J	- ELECTRONIC CONTROL MODULE CONNECTIONS
L	- INDICATOR LAMP
LFP	- LOW FUEL PRESSURE
MCB	- CIRCUIT BREAKER CONNECTIONS
MPU	- MAG. PICK UP
OI	- OPERATOR INTERFACE
OPS	- OIL PRESSURE SENDER
OTS	- OIL TEMP. SENDER
R	- RESISTOR
RB	- RELAY BOARD
RB_A	- RELAY BOARD CONNECTOR
SC	- START CONTACTOR
SM	- STARTER MOTOR CONNECTIONS
ST	- SHUNT TRIP RELAY
SW1	- AUTO/MANUAL SELECT SWITCH
TB	- TERMINAL BLOCKS
TG	- TRIP CONTACTOR RELAY BASE
WLS	- COOLANT LEVEL SENDER
WTS	- COOLANT TEMP. SENDER



Electrical Data

5.4L & 6.8L PM-PC Wiring Diagram - Drawing No. 0F2837-F Page 7





Generac Power Systems, Inc.

P.O. Box 8 • Waukesha, WI 53187

Phone: 262-544-4811